

# ENERGEX

## Pricing Proposal

for the period 1 July 2010–30 June 2011

May 2010



positive energy



positive energy

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# 1 Preface

The National Electricity Rules (the *Rules*) Section 6.18 requires a Distribution Network Service Provider (DNSP) to submit to the Australian Energy Regulator (AER) an initial pricing proposal for the first *regulatory year* of the *regulatory control period*, within 15 business days after publication of the distribution determination. This document has been prepared to comply with the requirements of the *Rules* and additional requirements as specified by the AER in its Final Determination.



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## 2 Introduction

### 2.1 Scope

This initial pricing proposal outlines the proposed network tariffs for *standard control services* and the approved prices for *alternative control services* for 2010-11. Network tariffs are based on the recovery of the maximum allowed revenue approved by the Australian Energy Regulator (AER) in the 2010 Final Determination for the *2010-15 regulatory control period*.

This document is submitted in accordance with, and complies with, the requirements of:

- The National Electricity Law (NEL);
- The National Electricity Rules (*the Rules*), Version 35 (25 March 2010);
- Final Determination, Queensland, 2010-11 to 2014-15 (AER, May 2010); and
- Framework and Approach Paper, Classification of services and control mechanisms, ENERGEX and Ergon Energy 2010-15 (AER, August 2008).

### 2.2 Structure

ENERGEX's initial pricing proposal is comprised of this proposal document and appendices and is structured as follows:

**Table 2.1 Structure of document**

| Chapter | Title                               | Purpose                                                                                                                                                                                                                                      |
|---------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2       | Introduction                        | Chapter 2 outlines the scope, structure and purpose of this initial pricing proposal.                                                                                                                                                        |
| 3       | Business characteristics            | Chapter 3 provides an overview of the ENERGEX business.                                                                                                                                                                                      |
| 4       | Pricing framework                   | Chapter 4 outlines the pricing principles and objectives which ENERGEX has followed in setting tariffs.<br><br>Chapter 4 also provides details of the modelling inputs and outputs used to develop tariffs to recover the regulated revenue. |
| 5       | Assignment of customers and tariffs | Chapter 5 outlines how customers and tariffs are assigned to tariff classes based on the <i>Rules</i> and pricing principles.                                                                                                                |
| 6       | Tariff classes – standard control   | Chapter 6 provides details of each tariff included under <i>standard control services</i> , including a description of each tariff class and the charging parameters which are related to each tariff.                                       |



| Chapter | Title                                         | Purpose                                                                                                                                                                                                                                                                |
|---------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7       | Tariff classes – alternative control services | Chapter 7 provides details of each tariff included under <i>alternative control services</i> , including a description of street lighting, fee based and quoted services (including large customer connections) and the charging parameters which are related to each. |
| 8       | Proposed tariff variations                    | Chapter 8 discusses the proposed variations to tariffs during the 2010-11 regulatory year and the aspects of tariffs which have changed from 2009-10 to 2010-11.                                                                                                       |
| 9       | Transmission cost recovery                    | Chapter 9 outlines how adjustments to charges for transmission use of system (TUOS) services resulting from over or under recovery are calculated.                                                                                                                     |
| 10      | Tariff development                            | Chapter 10 outlines tariff development in the medium term, including any additional tariffs, structural changes or removal of tariffs proposed over the <i>2010-15 regulatory control period</i> .                                                                     |
| 11      | Compliance with regulatory requirements       | Chapter 11 provides details of how this initial pricing proposal and the ENERGEX methodology complies with the requirements under the <i>Rules</i> and also under the requirements of the AER Final Determination.                                                     |
| 12      | Customer impacts                              | Chapter 12 provides a discussion on the customer impacts as a result of the proposed tariffs for the 2010-11 regulatory year.                                                                                                                                          |
| 13      | Annual approval process                       | Chapter 13 provides the relevant dates and details for the annual reporting and approvals.                                                                                                                                                                             |
| 14      | Glossary                                      | Chapter 14 defines the key terms and abbreviations used in this document.                                                                                                                                                                                              |
| 15      | Claim for Confidentiality                     | Chapter 15 details which parts of this document are confidential and provides reasons in support of a confidentiality claim.                                                                                                                                           |

## 2.3 Purpose

The purpose of this document is to describe the methodology and principles which ENERGEX has followed in proposing its tariffs to recover the \$1,084.5 million of revenue set by the AER for the first year of the *2010-15 regulatory control period* and to list the prices approved by the AER for *alternative control services*. This is ENERGEX's initial pricing proposal addressing the first year of the *regulatory control period*.



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Under the requirements of the *Rules*, the following applies:

### Regulatory Requirement

#### Clause 6.18.2 of the *Rules* states that:

- (a) A *Distribution Network Service Provider* must:
- (1) submit to the AER, as soon as practicable, and in any case within 15 business days, after publication of the distribution determination, a *pricing proposal* (the "initial *pricing proposal*") for the first *regulatory year* of the *regulatory control period*; and
  - (2) submit to the AER, at least 2 months before the commencement of the second and each subsequent *regulatory year* of the *regulatory control period*, a further *pricing proposal* (an "annual *pricing proposal*") for the relevant *regulatory year*.
- (b) A *pricing proposal* must:
- (1) set out the *tariff classes* that are to apply for the relevant *regulatory year*; and
  - (2) set out the proposed tariffs for each *tariff class*; and
  - (3) set out, for each proposed tariff, the *charging parameters* and the elements of service to which each *charging parameter* relates; and
  - (4) set out, for each *tariff class* related to *standard control services*, the expected weighted average revenue for the relevant *regulatory year* and also for the current *regulatory year*; and
  - (5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the *regulatory year* and the basis on which it could occur; and
  - (6) set out how charges incurred by the *Distribution Network Service Provider* for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous *regulatory year*; and
  - (7) demonstrate compliance with the *Rules* and any applicable distribution determination; and
  - (8) describe the nature and extent of change from the previous *regulatory year* and demonstrate that the changes comply with the *Rules* and any applicable distribution determination.



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In addition, as part of the Final Determination, the AER requested the following items to be included in ENERGEX's initial pricing proposal:

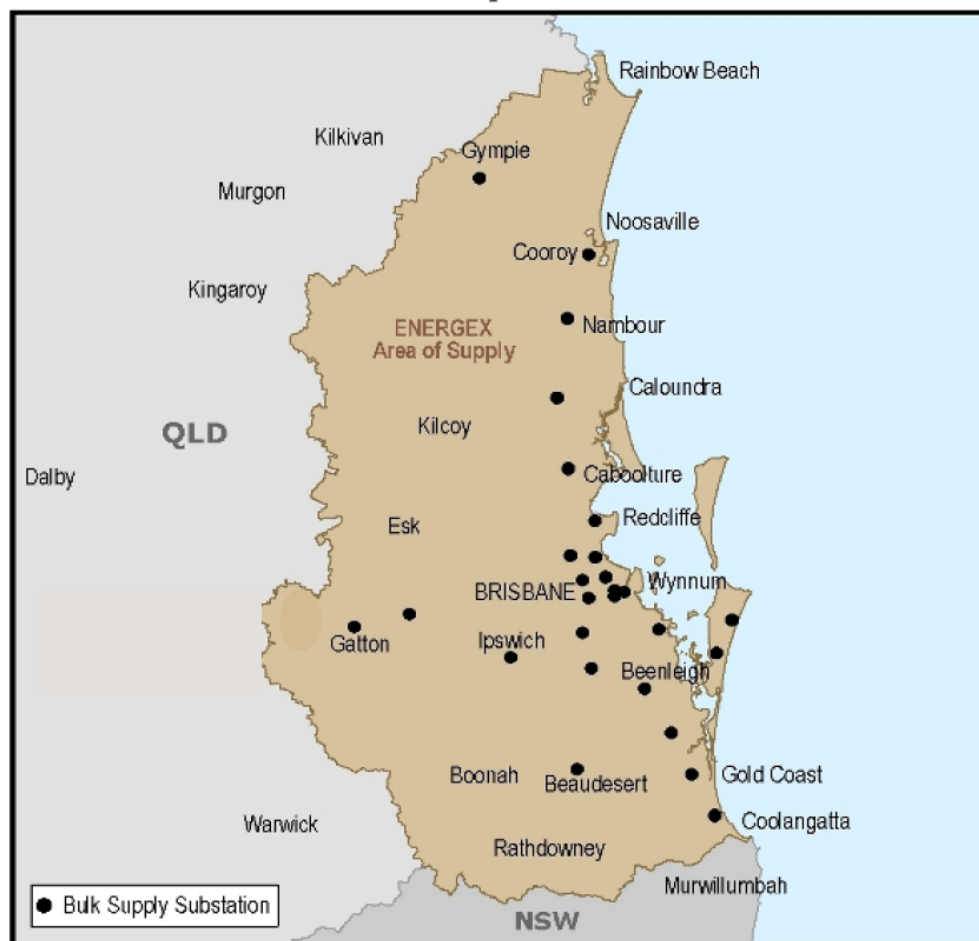
- in accordance with clause 6.12.1(19) of the *Rules*, a record of the amount of revenues recovered from transmission use of system (TUOS) charges and associated payments in accordance with Appendix E of the Final Determination;
- for street lighting, the proposed price for each service in the regulatory year and the revenues collected from the provision of each service in the preceding regulatory year;
- for fee based and quoted services, the volume of services provided and revenues collected from the provision of each service in the preceding regulatory year, in addition to the proposed price for each service in the regulatory year;
- information on a distribution use of system (DUOS) unders and overs account under clause 6.18.2(b)(7) of the *Rules*; and
- information on a TUOS unders and overs account under clause 6.18.2(b)(7) of the *Rules*.



### 3 Business characteristics

ENERGEX is a Brisbane-based company with over 3,700 employees and assets worth over \$8 billion<sup>1</sup>. ENERGEX's core business involves the distribution of electricity to more than 1.3 million residential, industrial and commercial connections across South East Queensland (SEQ). The company's electricity distribution network spans approximately 25,000 square kilometres throughout SEQ, stretching from Gympie in the north to Withcott in the west, Stradbroke Island in the east and Coolangatta in the south. This supply area encompasses the high growth regions of Brisbane, the Gold Coast and the Sunshine Coast.

**Figure 3.1 ENERGEX Area of Supply**



ENERGEX maintains assets which include more than 50,000 kilometres of underground cables and overhead lines, over half a million power poles, 250 zone and bulk supply substations, some 43,000 distribution transformers and approximately 300,000 street lights.

<sup>1</sup> Value as at 30 June 2009 and reported in ENERGEX's *Annual Report 2008-2009*, page 3.

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ENERGEX is authorised under the Electricity Act 1994 (Electricity Act) as a distribution entity with a supply area covering SEQ. ENERGEX's entire network is classified as a distribution network for the purposes of the *Rules*.

ENERGEX takes supply of electricity from Powerlink at transmission connection points and distributes the supply via the sub transmission and distribution network to customers throughout the region. Zone substations and distribution substations convert the voltages as necessary to minimise network losses and meet customers' voltage requirements. ENERGEX also operates some small distributed generation which supports the network during normal and contingency situations.

Providing electricity to a mixture of urban and rural areas, the ENERGEX electricity network is characterised by:

- high density/CBD areas such as the Brisbane CBD, Gold Coast and Sunshine Coast city areas which are supplied by 110/11 kV, 110/33 kV, 132/33 kV, or 132/11 kV injection points;
- suburban/urban/short rural feeder areas where 110/33 kV or 132/33 kV bulk supply substations are used to supply 33/11 kV zone substations;
- Brisbane suburban areas close to the CBD which have extensive older, meshed 33 kV underground cable networks that supply zone substations;
- outer suburbs and growth areas to the north, south and west of Brisbane which are supplied via modern indoor substations of modular design that enable further modules to be readily added;
- an increasing proportion of new network comprised of underground cables; and
- new subdivisions in urban/suburban areas which are supplied by underground networks with padmount substations.

ENERGEX is committed to the objectives of the NEL in relation to the management and operation of the electricity network in the long-term interests of customers with respect to price, quality, safety, reliability and security of supply.

As an electricity distribution business operating in SEQ, the current challenges for ENERGEX are meeting sustained demand growth, improving security and reliability, and renewal and replacement of assets within a changing operating environment.

To ensure its short-term objectives and strategies are met, ENERGEX is required to invest significant capital and operating expenditure in supply-side solutions during the *2010-15 regulatory control period*. At the same time, ENERGEX's longer term strategies will guide the business towards delivering sustainable future performance by improving the balance between supply-side management and demand-side solutions.



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## 4 Pricing Framework

### 4.1 Pricing Principles and Objectives

ENERGEX's overall objective in setting tariffs for *standard control services* is to ensure that the regulated revenue is recovered from customers in a manner that is consistent with the pricing principles as outlined in the *Rules*.

For *alternative control services* (subject to a price cap), the overall objective is to ensure that the price charged is cost reflective and is consistent with the pricing principles as outlined in the *Rules*.

The pricing principles listed below are applicable to *standard control services* and *alternative control services*.

#### Regulatory Requirement

**Clause 6.18.5 of the *Rules* states that:**

- (a) For each *tariff class*, the revenue expected to be recovered should lie on or between:
  - (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and
  - (2) a lower bound representing the avoidable cost of not serving those customers.
- (b) A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:
  - (1) must take into account the long run marginal cost for the service or, in the case of a *charging parameter*, for the element of the service to which the *charging parameter* relates; and
  - (2) must be determined having regard to:
    - (i) transaction costs associated with the tariff or each *charging parameter*; and
    - (ii) whether customers of the relevant *tariff class* are able or likely to respond to price signals.
- (c) If, however, as a result of the operation of paragraph (b), the *Distribution Network Service Provider* may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

In addition to the pricing principles established under the *Rules*, ENERGEX has applied the following pricing objectives in the formulation of tariffs. These additional objectives are intended to complement the requirements under the *Rules* and provide clarity in the formulation of robust tariffs.



- 
- **Free from cross subsidy** – for a network user, or group of users, there should be no cross subsidies between each tariff class of *standard control services*, or between standard control and alternative control tariffs;
  - **Efficient use of the network** – tariffs should incorporate appropriate signals to network users of their impact on existing and future network capacity and costs, and to encourage demand management;
  - **Equity** – tariffs should be equitable for customers and should reflect the users' utilisation of the existing network and the use of specific dedicated assets;
  - **Price stability** – tariffs should not widely fluctuate over time to permit customers to make informed investment decisions;
  - **Cost-reflectivity** – as far as possible, tariffs should reflect the actual cost of service provision to customers;
  - **Simplicity** – tariffs should be simple and straightforward to apply, based on a well-defined and clearly explained methodology and be readily understood by customers; and
  - **TUOS charges** – network tariffs should preserve the economic signals present in the structure of TUOS charges.

## 4.2 Setting the 2010-11 tariffs

This section provides a high level overview of how the regulated revenue is recovered through tariffs for *standard control services*. The tariffs which are set as a result of this process are provided in Section 6.

The 2010-11 tariffs for *alternative control services* are subject to a price cap form of control and are discussed in Section 7.

### 4.2.1 Maximum Allowable Revenue (MAR) and Revenue Cap

The maximum allowable revenue (MAR) formula which ENERGEX uses in determining the regulated revenue to recover is set by the AER in the Final Determination.

The 2010-11 tariffs and *charging parameters* set out in this initial pricing proposal lead to expected revenues (a revenue cap) consistent with the MAR plus any DUOS unders/overs adjustment from prior periods.

Table 4.1 details the revenue cap calculation for 2010-11. The revenue cap is based on a building block approach, which includes each of the regulated cost components, namely:

- regulatory depreciation;
- return on capital;
- operating expenditure; and
- tax allowance.



**Table 4.1 2010-11 Revenue Cap Calculations**

| Component                                                  | Amount (\$m)   | Comments/Reference                                                                 |
|------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|
| Regulatory Depreciation                                    | 78.5           | As approved in the May 2010 AER Queensland Final Determination 2010-11 to 2014-15. |
| Return on Capital                                          | 764.5          |                                                                                    |
| Operating Expenditure                                      | 325.6          |                                                                                    |
| Tax Allowance                                              | 32.6           |                                                                                    |
| Capital Contributions                                      | -65.1          |                                                                                    |
| Revenue from Shared Assets                                 | -4.0           |                                                                                    |
| <b>Annual Revenue Requirement</b>                          | <b>1,132.1</b> |                                                                                    |
| Smoothing / CPI                                            | 3.0            |                                                                                    |
| <b>Expected Revenue</b>                                    | <b>1,135.1</b> |                                                                                    |
| <b>Adjustments:</b>                                        |                |                                                                                    |
| Capital Contributions                                      | 0.9            | Adjustment for under recovery                                                      |
| Tax                                                        | -27.1          | Adjustment for over recovery                                                       |
| Service Target Performance Incentive Scheme (STPIS) Factor | 0.0            | No STPIS adjustment for 2010-11                                                    |
| Passthrough                                                | 0.0            | No approved passthrough amounts for 2010-11                                        |
| <b>MAR</b>                                                 | <b>1,108.9</b> | <b>Expected Revenue plus Total Adjustments</b>                                     |
| <b>Unders/Overs Adjustments:</b>                           |                |                                                                                    |
| DUOS                                                       | -24.4          | Adjustment for over recovery                                                       |
| <b>Revenue Cap</b>                                         | <b>1,084.5</b> | <b>MAR plus DUOS under/over adjustment</b>                                         |

#### 4.2.2 Modelling outputs

As outlined in Table 4.1, the total Revenue Cap for 2010-11 is \$1,084.5 million. The allowances for each of the regulatory cost components are adjusted to account for the smoothed revenue and the annual adjustments as per Table 4.2 below.



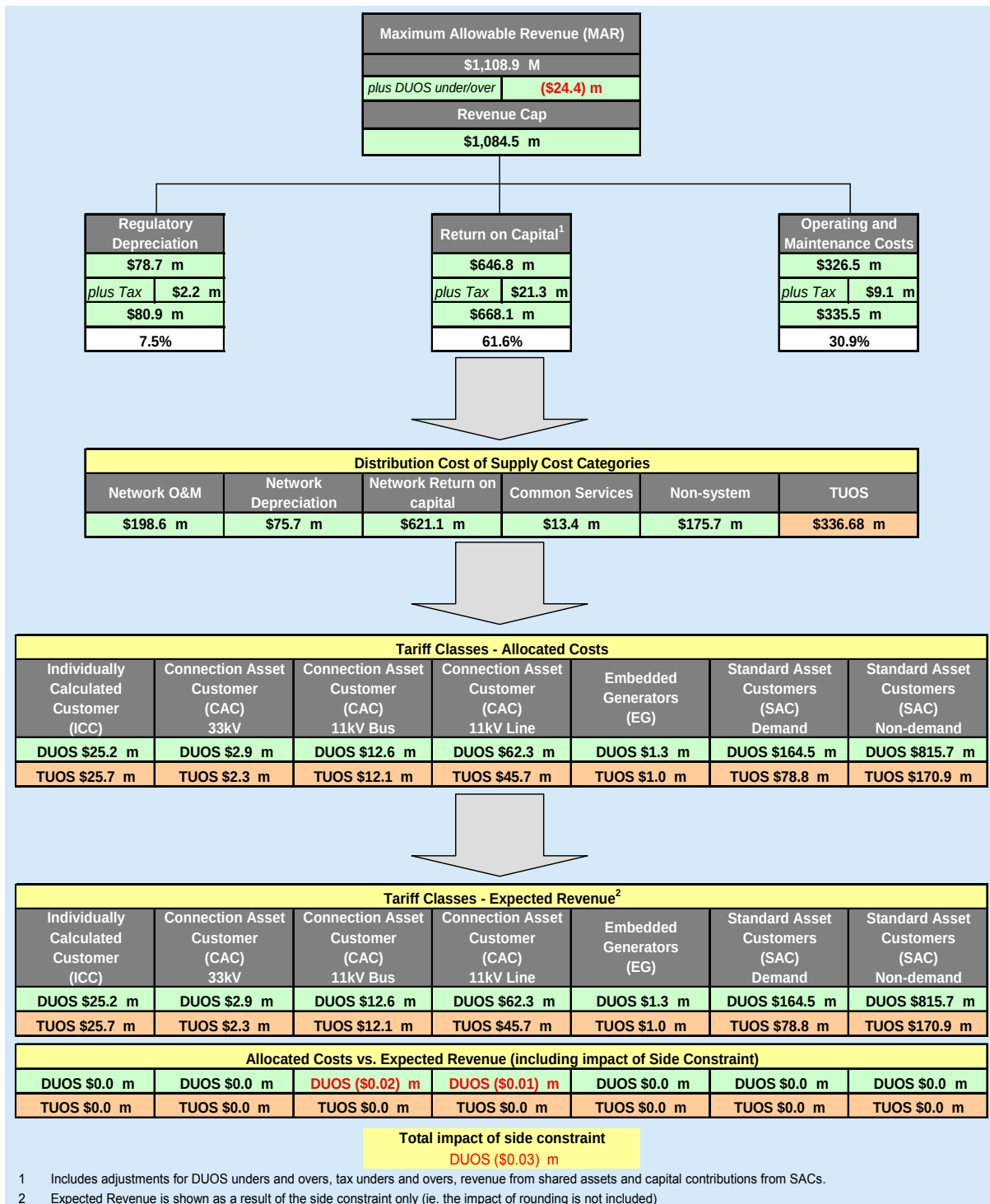
**Table 4.2    Redistribution of smoothed Revenue Cap to cost components**

| Component                                   | Sub Total<br>(\$m) | Total<br>(\$m) |
|---------------------------------------------|--------------------|----------------|
| <b>Regulatory Depreciation</b>              |                    |                |
| Original building block revenue requirement | 78.5               |                |
| Smoothing Allowance                         | 0.2                | 80.9           |
| Tax Allowance                               | 2.2                |                |
| <b>Return on Capital</b>                    |                    |                |
| Original building block revenue requirement | 764.5              |                |
| Smoothing Allowance                         | 1.9                |                |
| Tax Allowance                               | 21.3               |                |
| Capital Contributions                       | -65.1              |                |
| Revenue from Shared Assets                  | -4.0               | 668.1          |
| Unders/Overs Adjustments                    |                    |                |
| DUOS                                        | -24.4              |                |
| Capital Contributions                       | 0.9                |                |
| Tax                                         | -27.1              |                |
| <b>Operating Expenditure</b>                |                    |                |
| Original building block revenue requirement | 325.6              | 335.5          |
| Smoothing Allowance                         | 0.8                |                |
| Tax Allowance                               | 9.1                |                |
| <b>Revenue Cap</b>                          | <b>1084.5</b>      | <b>1084.5</b>  |

The first stage of the tariff determination process is to allocate or assign these network costs to the tariff classes in the most efficient and cost reflective way. ENERGEX allocates costs and tariffs to tariff classes using a Distribution Cost of Supply (DCOS) model. This modelling process is explained in Appendix 3 and the allocation of costs to recover the Revenue Cap is illustrated in Figure 4.1 below.



**Figure 4.1 2010-11 Tariff Development**



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### 4.2.3 Demand, Energy Forecasts and Customer Numbers

As part of the *Regulatory Information Notice* (RIN), ENERGEX provided the AER details of the demand and energy forecasts and expected customer numbers over the regulatory control period. The following figures for 2010-11 were approved by the AER in the Final Determination.

- Total Energy Consumption = 22,416 GW.h
- Total Customer Numbers = 1,363,138
- Maximum Demand = 4,931 MW

The energy consumption and customer number forecasts used to calculate 2010-11 tariffs both agree at the aggregate level to the energy consumption forecast of 22,416 GW.h and customer number forecast of 1,363,138 approved by the AER in the Final Determination.

The maximum demand shown above represents a forecast coincident system peak demand measured at the distribution network connections to the transmission system. However, ENERGEX uses anytime maximum demand (ATMD) as its demand cost allocator in the DCOS and pricing models. The sum of all ATMDs will not equal the system maximum demand shown above. This is because the ATMD is forecast from individual customer meter data and individual customers don't necessarily peak at the same time as the entire system (a concept known as diversity).

For pricing purposes ENERGEX has forecast the ATMD figures from actual customer meter data for 2009. This means pricing is calculated using forecasts based on the same data that was used for billing purposes. Where customers do not have meters capable of measuring demand, a forecast ATMD was derived from metered energy readings. This bottom-up demand forecast will exceed the system maximum demand forecast of 4,931 MW due to the diversity effect. However, the individual customer energy consumptions used in the calculation of 2010-11 tariffs will match the total energy consumption shown above.

Further discussion on the use of ATMD for pricing purposes is contained in Appendix 3.



---

# 5 Assignment of customers and tariffs

## 5.1 Standard control services

This section outlines ENERGEX's approach to the assignment and reassignment of customers to tariff classes to comply with Clause 6.18.4 of the *Rules*.

Each customer for *standard control services* is a member of at least one tariff class. ENERGEX's customer assignment to tariff classes is determined based on assessment of the following criteria:

- a) energy consumption and/or demand; and
- b) voltage level.

To provide the appropriate economic and cost of supply signals, the tariff regime distinguishes between different tariff classes. ENERGEX has established tariff classes (or customer groups) based on the criteria above, and which takes into account the relevant customers' capability to respond to price signals as well as their capacity to make investment and operational decisions that will affect their electricity usage profile.

In addition to the above, the following guidelines apply:

- allocation of a customer with micro-generation facilities to a tariff will be made on the same basis as other connections;
- where a new tariff is applied to a customer, it is standard practice to apply the tariff from the next billing period; and
- for new connections with no previous load history, the default tariff is based on their expected energy usage, supply voltage and meter type.

In accordance with Clauses 6.18.4(a)(4) and 6.18.4(b), assignment of customers to tariffs is reviewed periodically to assess if the tariff assignment is still applicable given potential changes in annual usage and meter type. A change in connection voltage means that the connection is treated as if it was a new connection.

To limit transaction costs and ensure pricing signals are not distorted by constant changes in customer tariff assignment, customers are generally only allowed one tariff change per 12 month period.

For customers with demand levels which fluctuate year to year, ENERGEX may apply a tolerance limit of up to 20 per cent around tariff thresholds to mitigate frequent tariff reassignment and subsequently limit customer impact.

The procedure for assigning and reassigning customers to tariff classes satisfies the requirements of Clause 6.18.4. This procedure relates specifically to the application of



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mandated tariffs. Customers who have chosen to be assigned to a voluntary tariff will not normally be subject to this review process.

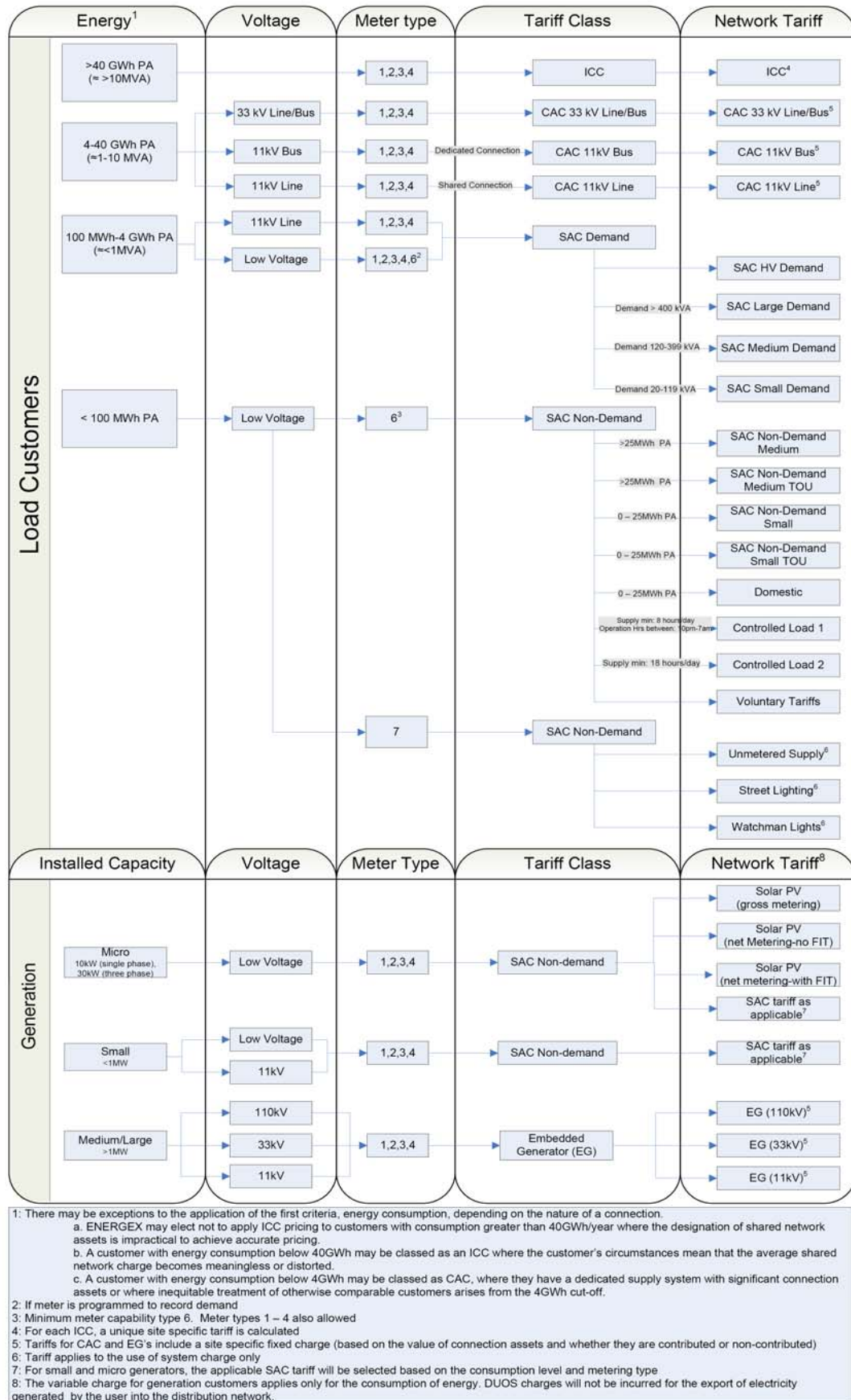
ENERGEX believes this approach satisfies the requirements of Clauses 6.18.3(d) and 6.18.4, namely:

- balancing the grouping of customers on an economically efficient basis with the need to avoid unnecessary transaction costs; and
- ensuring customers with a similar connection and usage profile are treated on an equal basis and customers with micro-generation facilities are not treated less favourably.

A pictorial representation of the process for assignment to tariff class for *standard control services* is outlined in Figure 5.1.



**Figure 5.1 Assignment of customers to tariff classes – *standard control services***



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### **5.1.1 Assignment of customers and dispute resolution process**

In accordance with the requirements of Appendix B of the Final Determination, the following process will be followed for the assignment or reassignment of customers to tariff classes:

1. ENERGEX will notify any customer concerned in writing of the tariff class to which the customer has been assigned or reassigned, prior to the assignment or reassignment occurring. If ENERGEX does not know the identity of the customer then the customer's retailer will be notified instead.
2. The notice given will include advice that the customer may request further information from ENERGEX and that it may object to the proposed assignment or reassignment. This notice will specifically include:
  - a. either a copy of ENERGEX's internal procedures for reviewing objections or the link to where such information is available on the ENERGEX website;
  - b. that if the objection is not resolved to the satisfaction of the customer under ENERGEX's internal review system, then to the extent that resolution of such disputes is within the jurisdiction of a state based energy ombudsman scheme the customer is entitled to escalate the matter to such a body; and
  - c. that if the objection is not resolved to the satisfaction of the customer under ENERGEX's internal review system, then the customer is entitled to seek resolution via the dispute resolution process available under Part 10 of the NEL.
3. If, in response to a notice issued in accordance with point 1, ENERGEX receives a request for further information from a customer, then ENERGEX will provide such information. If any of the information requested by the customer is confidential ENERGEX is not required to provide that information to the customer.
4. If, in response to a notice issued in accordance with point 1, a customer makes an objection to ENERGEX about the proposed assignment or reassignment, ENERGEX will reconsider the proposed assignment or reassignment, taking into consideration clause 6.18.4(a)(1) to (3) of the *Rules*, and will notify the customer in writing of its decision and the reasons for that decision.
5. If a customer's objection to a tariff assignment or reassignment is upheld by the relevant external dispute resolution body, then any adjustment which needs to be made to prices will be done by ENERGEX as part of the next annual review of prices.

### **5.1.2 Reassignment of customers in 2010-11**

The Final Determination states that each customer who was a customer of a Qld DNSP prior to 1 July 2010, and who continues to be a customer of a Qld DNSP as at 1 July 2010, will be taken to be "assigned" to the tariff class which the Qld DNSP was charging that customer immediately prior to 1 July 2010. This requirement will be followed, with the exception that, where applicable, ENERGEX will be reclassifying customers from 1 July 2010 in line with the *Rules* contained in clause 6.18.4, based on the periodic review process.

From 1 July 2010, a number of customers will be reassigned to a new tariff class on the basis of the customer's historical energy consumption. A complete list of the reclassifications for 1 July 2010 is provided in Appendix 4.



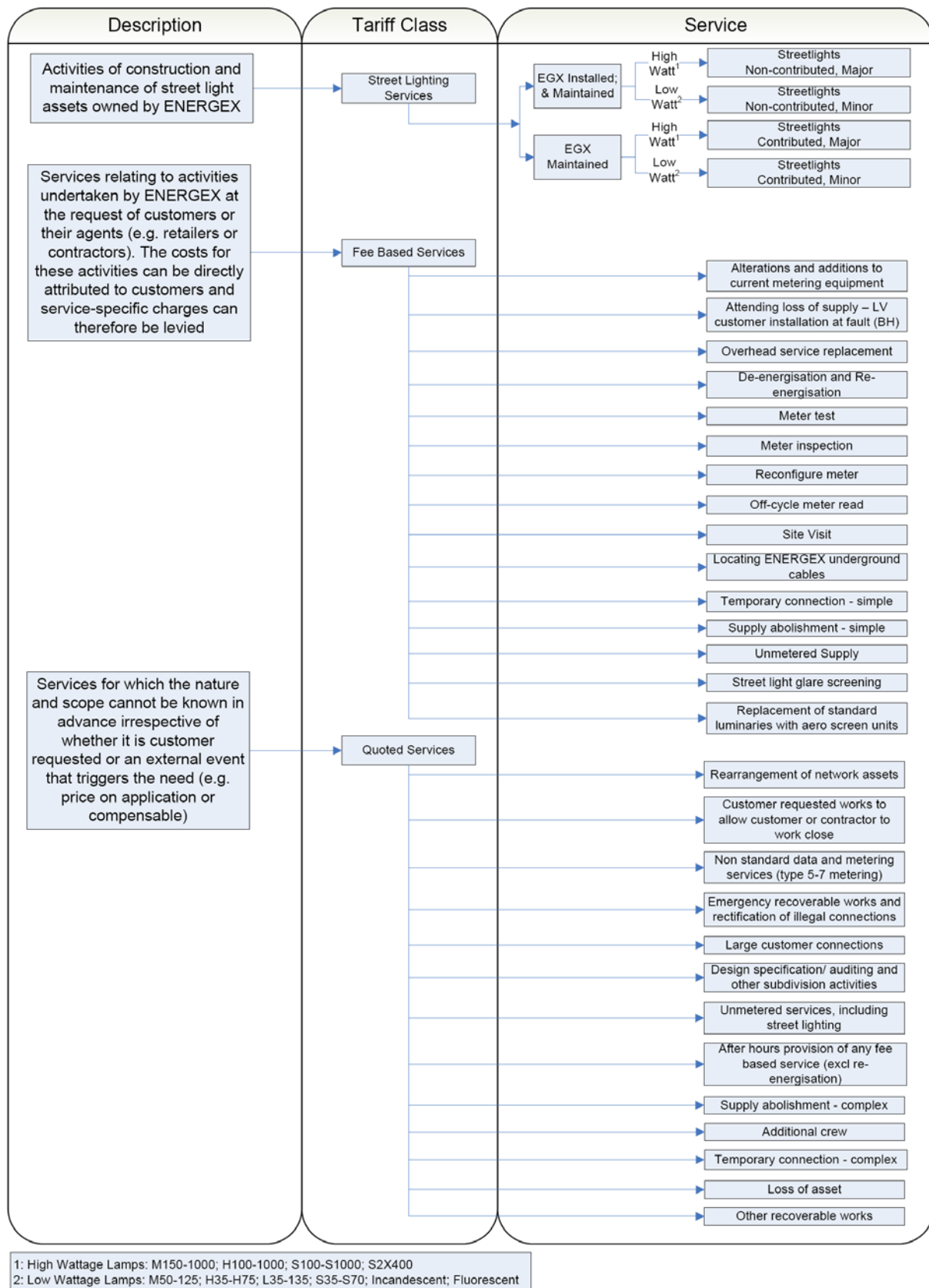
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## 5.2 Alternative control services

The assignment of customers to an *alternative control services* tariff is based on the type of service required. Each customer for *alternative control services* is a member of at least one tariff class. Figure 5.2 below outlines the tariff class and service types which are included under *alternative control services*.



**Figure 5.2 Assignment of customers to tariff classes – *alternative control services***



## 6 Tariff classes – standard control services

In the final framework and approach paper, the AER classified network services, connection services and metering services as *standard control services*. This initial pricing proposal is in accordance with the AER's decision.

### 6.1 Overview

In line with clause 6.18.3(d) of the *Rules*, tariff classes have been selected in recognition of the need to group together customers on an economically efficient basis which adequately reflects customer characteristics with regard to transaction costs. The tariff classes for *standard control services* are outlined in Table 6.1.

**Table 6.1 Tariff classes for *standard control services***

| Tariff Class | Tariff | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Tariff Schedule                                                                                                                                                                                                                                                                                                     |
|--------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ICC          | ICC    | <p>Typically those customers with electricity consumption greater than 40GW.h per year at a single connection point; or where a customer's circumstances mean that the average shared network charge becomes meaningless or distorted.</p> <p>Tariffs are based on actual TUOS charges for the relevant transmission connection point, plus charges associated with the customer's shared distribution network utilised for the electricity supply plus connection charges based on connection assets utilised.</p> <p>This provides the greatest cost reflectivity for these customers and is feasible since the number of such customers is relatively small. It is also justified by virtue of the shared distribution network assets being designated specifically to meet the requirements of these customers.</p> <p>Where there is a network on private property and there are site specific ENERGEX costs associated with operating, maintaining and accessing the network, these costs should be applied directly to the users of those assets when it is economically efficient to do so.</p> <p>The classification of customers as an ICC with electricity consumption lower than 40 GW.h per year at a single connection point could occur where:</p> <ul style="list-style-type: none"> <li>▪ A customer has a dedicated supply system which is quite different and separate from the remainder of the supply network;</li> <li>▪ There are only two or three customers in a supply system making average prices inappropriate; or</li> <li>▪ Inequitable treatment of otherwise comparable customers arises from the application of the 40 GW.h cut-off.</li> </ul> <p>Selection of these customers will be at ENERGEX's discretion.</p> | <p>The tariffs for ICCs are provided in Appendix 1.1 (Confidential).</p> <p>Tariffs are calculated on a site-specific basis and are not published due to the confidentiality requirements of the customer.</p> <p>ENERGEX will provide these site-specific tariffs directly to the customer and their retailer.</p> |



| Tariff Class            | Tariff                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Tariff Schedule                                                                                                                                                                                                                                                |
|-------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAC<br>33kV<br>Line/Bus | CAC<br>33kV<br>Line/Bus | Typically those customers with electricity consumption greater than 4GW.h (but less than 40GW.h) per year at a single connection point; or where a customer has a dedicated supply system with significant connection assets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | The site-specific components of tariffs for CACs are provided in Appendix 1.2 (Confidential).<br><br>ENERGEX will provide these site-specific tariffs directly to the customer and their retailer.<br><br>The other tariffs for CACs are outlined in Table 6.4 |
| CAC<br>11kV Line        | CAC<br>11kV Line        | These charges are based on average shared network charges plus site-specific connection charges based on the connection assets utilised, plus average TUOS charges.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                |
| CAC<br>11kV Bus         | CAC<br>11kV Bus         | <p>This provides a significant degree of cost reflectivity for this group of customers while recognising the practical difficulties of calculating individual shared network charges for each customer.</p> <p>Where there is a network on private property and there are site specific ENERGEX costs associated with operating, maintaining and accessing the network, these costs should be applied directly to the users of those assets when it is economically efficient to do so.</p> <p>The classification of customers as a CAC with electricity consumption lower than 4 GW.h per year at a single connection point could occur where:</p> <ul style="list-style-type: none"> <li>A customer has a dedicated supply system which is quite different and separate from the remainder of the supply network; or</li> <li>Inequitable treatment of otherwise comparable customers arises from the 4 GW.h per year cut-off.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                |
| Embedded Generator (EG) | EG                      | <p>The Embedded Generator (EG) tariff class relates generally to those generators with an installed capacity &gt;1MW in accordance with the Energy Networks Association (ENA) definition as follows:</p> <ul style="list-style-type: none"> <li>Medium: 1-5 MW (LV or HV) or &lt;1MW (HV); and</li> <li>Large: &gt;5MW.</li> </ul> <p>Tariffs for connection and access services for Medium and Large EGs will be developed on a similar basis to site-specific customers. This is due to the nature of connections, which are typically non-standard and may require additional embedded generator protection system upgrades.</p> <p>In accordance with the Rules, all EGs will receive a charge for connection services regardless of whether they are a net importer or exporter of electricity. However, DUOS charges will not be incurred for the export of electricity generated by the user into the distribution network. EGs who are net importers of electricity will receive appropriate volume charges.</p> <p>Micro and Small generators with an installed capacity &lt;1MW (LV) will be connected under an applicable SAC tariff based on the consumption level and metering type of the customer. However, a small generator may be treated as a CAC if there is a supply arrangement which is different to that which would be provided for a load customer of similar size or as per CAC requirements.</p> | <p>The site-specific components of tariffs for EGs are provided in Appendix 1.2 (Confidential).</p> <p>ENERGEX will provide these site-specific tariffs directly to the customer and their retailer.</p>                                                       |



| Tariff Class   | Tariff            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Tariff Schedule                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAC Demand     | HV Demand         | A Demand SAC tariff generally applies for those customers with consumption below 4GW.h per year, when the connection point has a meter installed that is capable of measuring energy consumption (kW.h) and demand (kW). This meter records total energy consumption and demand over 30 minute periods. A customer's demand is the average demand over the 30 minute period.<br><br>The tariff has been built to include a demand parameter so that the customer's actual demand is reflected in the price that they pay for their electricity. The highest demand reading for that month is used to calculate the customer's electricity bill.<br><br>Tariffs with a demand element are generally more cost reflective compared to those with only an energy charge.                                                                                                                                                            | Tariffs are based on the averages of the shared network prices and the connection prices for the customer.<br><br>The tariffs for SACs are outlined in Table 6.4.<br><br>Capital contributions apply primarily to newly connecting Standard Asset Customers (SACs), and are sought as prepayment for a revenue shortfall in the case of an uneconomic connection. ENERGEX's capital contributions policy is provided in Appendix 7. |
|                | Small Demand      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | Medium Demand     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | Large Demand      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| SAC Non-demand | Medium            | A Non-demand SAC tariff generally applies for customers with consumption below 4GW.h per year, when the customer's connection point has a meter installed that is capable of measuring total energy consumption (kW.h) only.<br><br>The tariff is selected based on the customer's annual consumption. The medium tariff applies to those customers with an annual consumption of 25MW.h or greater while the small tariff is applicable to those with an annual consumption less than 25MW.h.<br><br>Time-of-Use (ToU) tariffs take into account when, as well as how much, electricity is used by each consumer. With Time-of-Use, electricity is priced at multiple levels, depending on the time of day. Tariffs are lower during off-peak hours and higher during on-peak hours.                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | Medium TOU        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | Small             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | Small TOU         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | Domestic          | The domestic tariff generally applies to typical domestic customers with a consumption of up to 25MW.h per annum.<br><br>For domestic customers with consumption greater than 25MW.h, the Non-demand Medium tariff outlined above is applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | Solar PV          | The Solar Bonus Scheme is a program that pays domestic and other small energy customers for the surplus electricity generated from roof-top solar photovoltaic (PV) systems that is exported to the Queensland electricity grid. The solar bonus scheme applies to SACs who consume less than 100MW.h per annum. The Queensland Government sets the feed-in tariff rate (per kW.h) to be paid for the excess energy generated. ENERGEX pays this feed-in tariff.<br><br>In conjunction with the Solar Bonus Scheme, there are two other schemes that apply: <ul style="list-style-type: none"><li>Solar PV (net metering – no feed-in tariff) – where the electricity retailer purchases the excess energy generated. No feed-in tariff is provided by ENERGEX; and</li><li>Solar PV (gross metering) – where all energy generated is purchased by the electricity retailer. No feed-in tariff is provided by ENERGEX.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                | Controlled Load 1 | Specified permanently connected appliances are controlled by network equipment so that supply will be permanently available for a minimum period of 8 hours at the absolute discretion of ENERGEX but usually between the hours of 10:00pm and 7:00am.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Tariff Class               | Tariff            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                              | Tariff Schedule                                                                                                                                                   |
|----------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAC Non-demand (continued) | Controlled Load 2 | Specified connected appliances are controlled by network equipment so that supply will be available for a minimum period of 18 hours per day during time periods set at the absolute discretion of ENERGEX.                                                                                                                                                                                                                                              | Tariffs are based on the averages of the shared network prices and the connection prices for the customer.<br><br>The tariffs for SACs are outlined in Table 6.4. |
|                            | Other Unmetered   | The Other Unmetered supply tariff is applicable to unmetered supplies where the customer owns and maintains all assets after the customer connection point. This includes facilities such as public telephones, traffic signals, and public barbecues. ENERGEX only provides connection to the network for these services. The unmetered supply tariff therefore seeks to only recover a contribution towards the shared network (use of system charge). |                                                                                                                                                                   |
|                            | Street Lighting   | The provision, construction and maintenance of street lights are classified as an <i>alternative control service</i> from 1 July 2010. Hence, the standard control tariff seeks only to recover a contribution towards the shared network (use of system charge).                                                                                                                                                                                        |                                                                                                                                                                   |
|                            | Watchman Lights   | The capital and maintenance costs for the installation of watchman lights are classified as an unregulated service from 1 July 2010. Hence, the standard control tariff seeks only to recover a contribution towards the shared network (use of system charge).                                                                                                                                                                                          |                                                                                                                                                                   |

## 6.2 Charging parameters

### Regulatory Requirement

#### Clause 6.18.2(b)(3) of the *Rules* states that:

A *pricing proposal* must: set out, for each proposed tariff, the *charging parameters* and the elements of service to which each *charging parameter* relates.

Consistent with the *Rules*, the tariffs proposed by ENERGEX will be comprised of a number of charging parameters, each selected and structured to provide signals to customers about the efficient use of the network and their impact on future network capacity and costs.

The tariff structure and the proportioning of charge parameters have been developed to achieve the pricing principles and objectives listed previously. In developing the tariffs, ENERGEX has sought to select tariff parameters which signal the impact that the end users will have on the network whilst managing the demand and volume variance risk, minimising boundary issues between and within tariff classes and avoiding any signals that may result in perverse outcomes.

ENERGEX provides for the following charging parameters in the tariffs:

- **Fixed or standing charge parameters** are intended to reflect the incremental costs that arise from the connection and management of the customer. This sends a clear signal about the economic value of the dedicated connection and provides a stable price and income regardless of the usage of the network. Network usage signals are provided by other charging parameters.



- 
- **Demand charge parameters** are reflective of augmentation costs associated with customer demand activity. There are two demand charge parameter types:
    - a *monthly maximum demand charge* is levied on the basis that network users who place greater pressure on the system should incur higher charges. Network expansion becomes necessary where there is a likelihood of demand exceeding available capacity. While this demand fluctuates over time, the critical supply level to be provided is the aggregated network usage during peak times. Peak demand is the driving factor behind system augmentation and network users are charged according to their contribution to this threshold demand level; and
    - a *capacity charge* is similar to a monthly maximum demand charge, but more effectively assigns an adequate share of costs associated with system augmentations to low load factor network users, such as those who make maximum demands on the system at infrequent intervals. It sends signals to these customers that they can reduce their network charges by reducing the variability of their consumption, as this lower peak demand reduces the pressure that they place on the network.

Demand charges account for augmentation costs at the customer connection level and all associated upstream augmentation costs. For example, if a network user is connected at the low voltage level, the ability of the system to meet an increase in that user's demand is dependent upon the available capacity of assets further up the distribution network – low voltage, distribution, zone substations and sub-transmission.

Potentially, customers connected at a lower voltage distribution level, will require a greater capital expenditure for supply (due to a greater proportion of upstream costs), and the greater will be the demand charge levied.

- **Volume charge parameters** have been adopted to recover those costs that are related to the size of the customer but not specifically the demand of that customer on the network. These costs include overheads and common service costs. Volume charges are also used to recover any residual allocated costs after the capacity and demand charges have been calculated. For non-demand customers the volume charge parameter also recovers those costs that would have been recovered from a demand based charge. A volume charge parameter can be structured either as flat anytime energy charge or a Time of Use energy charge. By using a Time of Use structure, a volume charge can provide a small price signal that encourages efficient use of the network.

### 6.2.1 Recovery of DUOS

Network tariffs and charging parameters are designed to recover the approved revenue which is consistent with the calculation of the Maximum Allowable Revenue (MAR) as set out in the Final Determination. The network charging parameters adopted by ENERGEX for the recovery of DUOS for *standard control services* tariffs are detailed in Table 6.2.



**Table 6.2 Tariff charging parameters for DUOS charges**

| Tariff Class        | Tariff              | Tariff Charging Parameter                                     |                                               |                                                           |                                                                  |                                     |
|---------------------|---------------------|---------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------|-------------------------------------|
|                     |                     | Fixed or Service Availability Charge (\$/period)              | Capacity Charge <sup>1,2</sup> (\$/kW/period) | Monthly Maximum Demand Charge <sup>1</sup> (\$/kW/period) | Power Factor Adjustment (PFA) for Capacity & Demand <sup>3</sup> | Volume Charge <sup>4</sup> (¢/kW.h) |
| ICC                 | ICC                 | ✓                                                             | ✓                                             | ✓                                                         | ✓                                                                | ✓                                   |
| CAC – 33kV Line/Bus | CAC – 33kV Line/Bus | ✓                                                             | ✓                                             | ✓                                                         | ✓                                                                | ✓                                   |
| CAC – 11kV Line     | CAC – 11kV Line     |                                                               |                                               |                                                           |                                                                  |                                     |
| CAC – 11kV Bus      | CAC – 11kV Bus      |                                                               |                                               |                                                           |                                                                  |                                     |
| EG                  | EG                  | ✓                                                             | ✓                                             | ✓ <sup>5</sup>                                            |                                                                  | ✓ <sup>5</sup>                      |
| SAC Demand          | HV Demand           | ✓                                                             |                                               | ✓                                                         |                                                                  | ✓                                   |
|                     | Small Demand        |                                                               |                                               |                                                           |                                                                  |                                     |
|                     | Medium Demand       |                                                               |                                               |                                                           |                                                                  |                                     |
|                     | Large Demand        |                                                               |                                               |                                                           |                                                                  |                                     |
| SAC Non-demand      | Medium              | ✓                                                             |                                               |                                                           |                                                                  | ✓                                   |
|                     | Medium TOU          | ✓                                                             |                                               |                                                           |                                                                  | ✓                                   |
|                     | Small               | ✓                                                             |                                               |                                                           |                                                                  | ✓                                   |
|                     | Small TOU           | ✓                                                             |                                               |                                                           |                                                                  | ✓                                   |
|                     | Domestic            | ✓                                                             |                                               |                                                           |                                                                  | ✓                                   |
|                     | Solar PV            | Not applicable (Solar PV is subject to a feed in tariff only) |                                               |                                                           |                                                                  |                                     |
|                     | Controlled Load 1   | ✓                                                             |                                               |                                                           |                                                                  | ✓                                   |
|                     | Controlled Load 2   | ✓                                                             |                                               |                                                           |                                                                  | ✓                                   |
|                     | Unmetered           |                                                               |                                               |                                                           |                                                                  | ✓                                   |
|                     | Street lighting     |                                                               |                                               |                                                           |                                                                  | ✓                                   |
|                     | Watchman lights     |                                                               |                                               |                                                           |                                                                  | ✓                                   |

1. The application of capacity or demand-based charges (either authorised or actual) is limited by the type of metering installed. Demand charges are not appropriate for those customers with metering equipment only capable of measuring and recording delivered electricity volume.
2. In general, the capacity charge is levied on the basis of either contracted demand or the maximum demand in the previous calendar year.
3. From 1 July 2010, charges for ICCs and CACs will be adjusted up or down by a relative amount based on the difference between the customer's measured power factor and the power factor neutral point. The adjustment will be applied to the demand and capacity charge elements.
4. Volume charge may be either or a combination of flat or ToU.
5. These rates apply to EGs only when they are importing electricity to their site (i.e. as a consumer).



## 6.2.2 Recovery of TUOS

Electricity is delivered to ENERGEX's network from the generators via Powerlink's transmission network. ENERGEX pays TUOS charges to Powerlink on behalf of its customers and recovers these costs through the network tariffs.

ENERGEX's transmission cost recovery tariffs are based on a forecast of TUOS charges for each year, adjusted for under or over recoveries. The TUOS charges from Powerlink comprise both fixed and variable charges. Where administratively efficient and locational signals are material, the forecast TUOS charges will be passed on to customers in the same form of price structure as received from Powerlink. Powerlink is planning to move towards a greater proportion of demand based charges. As Powerlink make changes to their price structure, ENERGEX will endeavour to reflect these in the TUOS tariffs.

A forecast of the TUOS charges is provided by Powerlink in March of each year to allow ENERGEX to develop tariff components to recover the anticipated TUOS costs.

The network charging parameters adopted by ENERGEX for TUOS services are detailed in Table 6.3.

**Table 6.3 Tariff charging parameters for TUOS charges**

| Tariff Class        | Tariff              | Tariff Charging Parameter                        |                                               |                                   |                                                           |                                     |
|---------------------|---------------------|--------------------------------------------------|-----------------------------------------------|-----------------------------------|-----------------------------------------------------------|-------------------------------------|
|                     |                     | Fixed or Service Availability Charge (\$/period) | Capacity Charge <sup>1,2</sup> (\$/kW/period) | Common Service Charge (\$/period) | Monthly Maximum Demand Charge <sup>1</sup> (\$/kW/period) | Volume Charge <sup>3</sup> (¢/kW.h) |
| ICC                 | ICC                 | ✓                                                | ✓                                             | ✓                                 |                                                           | ✓                                   |
| CAC – 33kV Line/Bus | CAC – 33kV Line/Bus | ✓                                                | ✓                                             |                                   |                                                           | ✓                                   |
| CAC – 11kV Line     | CAC – 11kV Line     |                                                  |                                               |                                   |                                                           |                                     |
| CAC – 11kV Bus      | CAC – 11kV Bus      |                                                  |                                               |                                   |                                                           |                                     |
| EG                  | EG                  | ✓                                                | ✓                                             |                                   |                                                           | ✓ <sup>4</sup>                      |
| SAC Demand          | HV Demand           |                                                  |                                               |                                   | ✓                                                         | ✓                                   |
|                     | Small Demand        |                                                  |                                               |                                   |                                                           |                                     |
|                     | Medium Demand       |                                                  |                                               |                                   |                                                           |                                     |
|                     | Large Demand        |                                                  |                                               |                                   |                                                           |                                     |



| Tariff Class                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Tariff            | Tariff Charging Parameter                                     |                                               |                                   |                                                           |                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------|-----------------------------------------------|-----------------------------------|-----------------------------------------------------------|-------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   | Fixed or Service Availability Charge (\$/period)              | Capacity Charge <sup>1,2</sup> (\$/kW/period) | Common Service Charge (\$/period) | Monthly Maximum Demand Charge <sup>1</sup> (\$/kW/period) | Volume Charge <sup>3</sup> (¢/kW.h) |
| SAC Non-demand                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Medium            | ✓                                                             |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Medium TOU        | ✓                                                             |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Small             | ✓                                                             |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Small TOU         | ✓                                                             |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Solar PV          | Not applicable (Solar PV is subject to a feed-in tariff only) |                                               |                                   |                                                           |                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Domestic          | ✓                                                             |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Controlled Load 1 | ✓                                                             |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Controlled Load 2 | ✓                                                             |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Unmetered         |                                                               |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Street lighting   |                                                               |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Watchman lights   |                                                               |                                               |                                   |                                                           | ✓                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |                                                               |                                               |                                   |                                                           |                                     |
| <p>1 The application of capacity or demand-based charges (either authorised or actual) is limited by the type of metering installed. Demand charges are not appropriate for those customers with metering equipment only capable of measuring and recording delivered electricity volume.</p> <p>2 In general, the capacity charge is levied on the basis of either contracted demand or the maximum demand in the previous calendar year.</p> <p>3 Volume charge may be either or a combination of flat or ToU.</p> <p>4 These rates apply to EGs only when they are importing electricity to their site (that is as a load customer).</p> |                   |                                                               |                                               |                                   |                                                           |                                     |

## 6.3 Tariffs

### Regulatory Requirement

**Clause 6.18.2(b)(2) of the Rules states that:**

*A pricing proposal must: set out the proposed tariffs for each tariff class.*

The proposed tariffs for *standard control services* are provided in Table 6.4.

Site specific tariffs for ICC, CAC and EG customers are provided in Appendix 1. The tariffs for *standard control services* are set at the beginning of the regulatory year. However, within a regulatory year there may sometimes be a requirement to include a new site specific charge. The procedure in Appendix 6 describes the process for the approval of a new or revised site specific price for large customers.



**Table 6.4 Proposed tariffs for DUOS and TUOS – *standard control services***

| Tariff Class        | Tariff Description  | Network<br>Tariff<br>Code                                                   | DUOS Charges                                 |                               |                             |                   |                           |                               | TUOS Charges                                 |                               |                             |                   |                           |                               |
|---------------------|---------------------|-----------------------------------------------------------------------------|----------------------------------------------|-------------------------------|-----------------------------|-------------------|---------------------------|-------------------------------|----------------------------------------------|-------------------------------|-----------------------------|-------------------|---------------------------|-------------------------------|
|                     |                     |                                                                             | Fixed<br>(\$/day)                            | Capacity<br>(\$/kW/<br>month) | Demand<br>(\$/kW/<br>month) | Energy<br>(c/kWh) | Peak<br>Energy<br>(c/kWh) | Off-Peak<br>Energy<br>(c/kWh) | Fixed<br>(\$/day)                            | Capacity<br>(\$/kW/<br>month) | Demand<br>(\$/kW/<br>month) | Energy<br>(c/kWh) | Peak<br>Energy<br>(c/kWh) | Off-Peak<br>Energy<br>(c/kWh) |
| ICC                 | ICC                 | Tariffs for ICC customers are confidential and are provided in Appendix 1.1 |                                              |                               |                             |                   |                           |                               |                                              |                               |                             |                   |                           |                               |
| CAC – 33kV Line/Bus | CAC – 33kV Line/Bus | 3500                                                                        | Site<br>specific<br>(see<br>Appendix<br>1.2) | 0.55579                       | 1.42214                     |                   | 0.177                     | 0.018                         | Site<br>specific<br>(see<br>Appendix<br>1.2) | 0.40515                       |                             |                   | 0.663                     | 0.663                         |
| CAC – 11kV Bus      | CAC – 11kV Bus      | 4000                                                                        |                                              | 1.07489                       | 2.41804                     |                   | 0.226                     | 0.023                         |                                              | 0.56136                       |                             |                   | 0.618                     | 0.618                         |
| CAC – 11kV Line     | CAC – 11kV Line     | 4500                                                                        |                                              | 2.04251                       | 4.20031                     |                   | 0.289                     | 0.029                         |                                              | 0.54400                       |                             |                   | 0.577                     | 0.577                         |
| EG                  | EG                  | Tariffs for EG customers are confidential and are provided in Appendix 1.2  |                                              |                               |                             |                   |                           |                               |                                              |                               |                             |                   |                           |                               |
| SAC Demand          | HV Demand           | 8000                                                                        | 31.94563                                     |                               | 6.80532                     | 0.110             |                           |                               | 8.18090                                      |                               | 1.08302                     | 1.211             |                           |                               |
|                     | Large Demand        | 8100                                                                        | 32.17693                                     |                               | 7.65275                     | 0.110             |                           |                               | 5.85585                                      |                               | 1.19036                     | 1.236             |                           |                               |
|                     | Medium Demand       | 8200                                                                        | 12.53219                                     |                               | 9.05404                     | 0.112             |                           |                               | 4.72174                                      |                               | 1.25269                     | 1.238             |                           |                               |
|                     | Small Demand        | 8300                                                                        | 1.13298                                      |                               | 14.13322                    | 0.323             |                           |                               | 1.56126                                      |                               | 1.40135                     | 0.888             |                           |                               |
| SAC Non-demand      | Medium              | 8600                                                                        | 0.61949                                      |                               |                             | 6.144             |                           |                               | 0.51356                                      |                               |                             | 1.344             |                           |                               |
|                     | Medium TOU          | 8800                                                                        | 0.61949                                      |                               |                             |                   | 6.681                     | 4.009                         | 0.51356                                      |                               |                             |                   | 1.344                     | 1.344                         |
|                     | Small               | 8500                                                                        | 0.22063                                      |                               |                             | 6.680             |                           |                               | 0.05409                                      |                               |                             | 1.331             |                           |                               |
|                     | Small TOU           | 8700                                                                        | 0.22063                                      |                               |                             |                   | 6.814                     | 4.088                         | 0.05409                                      |                               |                             |                   | 1.331                     | 1.331                         |
|                     | Solar PV            | Government mandated tariff paid by ENERGEX to the customer.                 |                                              |                               |                             |                   |                           |                               |                                              |                               |                             |                   |                           |                               |
|                     | Domestic            | 8400                                                                        | 0.22063                                      |                               |                             | 6.680             |                           |                               | 0.05409                                      |                               |                             | 1.331             |                           |                               |
|                     | Controlled Load 1   | 9000                                                                        | 0.11139                                      |                               |                             | 0.348             |                           |                               | 0.00220                                      |                               |                             | 0.087             |                           |                               |
|                     | Controlled Load 2   | 9100                                                                        | 0.11819                                      |                               |                             | 1.256             |                           |                               | 0.00206                                      |                               |                             | 0.102             |                           |                               |
|                     | Street Lighting     | 9400                                                                        |                                              |                               |                             | 4.398             |                           |                               |                                              |                               |                             | 1.507             |                           |                               |
|                     | Watchman Lights     | 9500                                                                        |                                              |                               |                             | 4.398             |                           |                               |                                              |                               |                             | 1.507             |                           |                               |
|                     | Unmetered           | 9600                                                                        |                                              |                               |                             | 4.398             |                           |                               |                                              |                               |                             | 1.507             |                           |                               |

Note: All tariffs are GST exclusive



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## 7 Tariff classes – alternative control services

In the final framework and approach paper, the AER classified fee based services, quoted services and street lighting, as *alternative control services*. These three tariff classes are described in Table 7.1.

**Table 7.1** Tariff classes for *alternative control services*

| Tariff Class       | Definition                                                                                                                                                                                                                                                      |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Street Lighting    | Street lighting services relate to activities of provision, construction and maintenance of street light assets owned by ENERGEX (conveyance of electricity for street lights remains a <i>standard control service</i> ).                                      |
| Fee Based Services | Services relating to activities undertaken by ENERGEX at the request of customers or their agents (e.g. retailers or contractors). The costs for these activities can be directly attributed to customers and service-specific charges can therefore be levied. |
| Quoted Services    | Services for which the nature and scope cannot be known in advance irrespective of whether it is customer requested or an external event that triggers the need (e.g. price on application or compensable).                                                     |

### 7.1 Street lighting

This section describes ENERGEX's approach to the provision of the *alternative control service* of street lighting for the *2010-15 regulatory control period*.

#### 7.1.1 Overview

From 1 July 2010, the bundled prices that customers are charged for street lighting will consist of a *standard control service* and an *alternative control service*. The conveyance of electricity to street lights remains a *standard control service*, while services relating to the provision, construction and maintenance of street lighting assets have been classified by the AER as *alternative control services*. It is this *alternative control services* element of street lighting services that is discussed below.

In the *2010-15 regulatory control period*, ENERGEX will continue to provide for non-contributed and contributed street lighting services. From 1 July 2010, the non-contributed service will only apply where ENERGEX has constructed standard street lights. Customers will still receive an ongoing charge for the standard level of service provided under the non-contributed service. Non-standard street lights will be available as a contributed service or as a quoted service, charged as an incremental cost to a standard street light. Charges associated with these services will need to be paid upfront by the customer. In the instances



where work is required outside of business hours due to maintenance access restrictions or customer requirements, these services will be provided as a quoted service.

The AER has decided a price cap form of control for street lighting services will apply over the *2010-15 regulatory control period*. This consists of:

- a schedule of fixed prices for street lighting services for 2010-11; and
- a price path for the remaining years of the *2010-15 regulatory control period*.

ENERGEX's street lighting services for the *2010-15 regulatory control period* are outlined in the Table 7.2. ENERGEX provides major and minor street lighting services. The classification of major and minor refers to the road categorisation where the street light is located, which is a significant driver of costs (e.g. lamp size and associated installation and maintenance costs). For major or minor street lighting services, the network charge applicable (i.e. non-contributed or contributed) depends on whether the street light is funded by ENERGEX, or the customer or their agent.

**Table 7.2 Outline of *alternative control services* – street lighting for the *2010-15 regulatory control period***

| Street lighting service | Located on main or arterial road | Non-contributed service <sup>1</sup> | Contributed service <sup>2</sup> |
|-------------------------|----------------------------------|--------------------------------------|----------------------------------|
| Major                   | Yes                              | Yes                                  | Yes                              |
| Minor                   | No                               | Yes                                  | Yes                              |

1 Non-contributed service – Network charge includes cost of supply (capital), installation and maintenance. Street light is owned by ENERGEX.

2 Contributed service – Network charge includes cost of maintenance. Supply (capital) and installation of the street light is funded by the customer or their agent. Ownership is vested in ENERGEX on completion of the installation and upon commissioning.

### 7.1.2 Framework

The approach for the treatment of street lighting assets, contributed and non-contributed, will satisfy NER requirements and deliver network charges which directly correlate with the level of service provided.

#### 7.1.2.1 Street lighting services where ENERGEX constructs the asset

In the *2010-15 regulatory control period* where a non-standard street light is requested by the customer, a separate charge for the incremental cost difference will be levied as a quoted service. The customer will still receive an ongoing charge for the non-contributed service.

#### 7.1.2.2 Street lighting services where customer or agent constructs the asset

In the *2010-15 regulatory control period* ENERGEX proposes to recognise gifted assets as contributed assets and not seek to recover any asset related costs from the customer. The



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customer will receive an ongoing charge for the maintenance of the street light under the contributed service.

#### 7.1.2.3 Existing Contributed

ENERGEX will account for the residual contributed, but recognised, street light assets within *standard control services*. This aligns with the historical capital contributions treatment where benefits from the street light asset contributions were accrued by all SACs. This reflects that all SACs are the end-beneficiary of the contributed street lighting service.

### 7.1.3 Charging parameters

#### Regulatory Requirement

**Clause 6.18.2(b)(3) of the *Rules* states that:**

*A pricing proposal must set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.*

ENERGEX's revenue requirements for street lighting services have been determined based on the revenue building block components consistent with the approach used for its *standard control services*. Based on the revenue requirement produced from the Post Tax Revenue Model (PTRM), ENERGEX has developed prices for the first year of the *regulatory control period* using a methodology that applies a revenue allocation based on the relative installation costs for major and minor street lights and the applicable asset funding arrangement (non-contributed and contributed).

Street lights are allocated to major and minor according to luminaire type and size, and to non-contributed and contributed based on the funding arrangement.

This methodology is outlined in the following sections:

#### 7.1.3.1 Connection charge for non-contributed street lighting services

The revenue requirement for the recovery of forecast connection costs equates to the return on capital and regulatory depreciation for non-contributed assets. The connection charge revenue is apportioned to major and minor street lighting services based on the relative installation costs for a typical street light configuration for the relevant locality (i.e. major or minor road). The relevant proportion is derived from replacement costs<sup>2</sup> for a sample of commonly used street light configurations of luminaire, pole type and outreach bracket, weighted by the forecast number of street lights. The proportions applied from 2010-11 are set at 47 per cent for major services and 53 per cent for minor services.

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<sup>2</sup> Replacement costs were updated as at 28 April 2009 and the pricing methodology has been updated accordingly. Pricing methodology inputs are reviewed annually and monitored for appropriateness.



### 7.1.3.2 Operating charge for all street lighting services

The revenue requirement for the recovery of forecast operating expenditure is apportioned to:

- **major and minor street lighting services** – based on the same proportions as used for the connection charge; and
- **non-contributed and contributed services** – based on the proportion of forecast street lights under the respective funding arrangements.

In determining the operating charge in 2010-11, the proportions in Table 7.3 have been applied.

**Table 7.3 Proportions allocated for street light service operating charges**

| Street lighting service | Revenue proportion | Tariff          | Revenue proportion |
|-------------------------|--------------------|-----------------|--------------------|
| Major                   | 47%                | Non-contributed | 48%                |
|                         |                    | Contributed     | 52%                |
| Minor                   | 53%                | Non-contributed | 50%                |
|                         |                    | Contributed     | 50%                |

This methodology for calculating the target revenue for the respective charges is considered to provide a balance between cost reflective pricing, simplicity and administrative costs.

The formula to calculate the individual tariff is outlined below:

$$\frac{\text{Annual Target Revenue for Street Light Tariff}}{\text{Number of lamps for Street Light Tariff}} \div \text{Days in the year}$$

### 7.1.4 Tariffs

#### Regulatory Requirement

**Clause 6.18.2(b)(2) of the Rules states that:**

*A pricing proposal must set out the proposed tariffs for each tariff class.*

**The Final Determination from the AER requires:**

ENERGEX to provide for street lighting, the proposed price for each service in the *regulatory year* and the revenues collected from the provision of each service in the preceding *regulatory year*.

ENERGEX has provided a price schedule for the provision, construction and maintenance of street lights for the *2010-15 regulatory control period* in Appendix 2.1. The prices are based on the methodology outlined above and charges are tailored to enable the customer to be charged according to the level of service requested. The prices reflect standardised lights



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and no restriction on access for operation, maintenance and repair. An additional charge may apply in the case of restricted access. Prices for the subsequent years of the *2010-15 regulatory control period* have been developed based on 2010-11 prices and the proposed price path.

In line with the AER requirements, the revenues collected from the provision of each service in the preceding regulatory year are provided in Appendix 2.4.

The applicable terms and conditions for each street light service are outlined in ENERGEX's published price schedule and are based on the following principles:

- the contributed street light tariff only applies where the capital cost of the street light has been paid upfront by the customer or their agent;
- at the conclusion of the street light's standard asset life, the non-contributed tariff will apply;
- where ENERGEX has supplied the street light and the capital cost has been paid upfront by the customer, the contributed street light tariff will apply for the minimum period specified in the tariff's terms and conditions; and
- where the capital cost of the street light has been funded by ENERGEX and no upfront payment made, whether the initial asset construction or replacement of an asset, the non-contributed street light tariff applies.

## 7.2 Fee based and quoted services

This section describes ENERGEX's approach to the provision of the *alternative control service* of fee based and quoted services for the *2010-15 regulatory control period*.

### 7.2.1 Overview

ENERGEX's other *alternative control services* relate to the fee-based services and quoted services, including large customer connections, provided by ENERGEX which are ancillary to the main distribution services and are provided at the explicit request of third parties.

- **Fee-based services** – are services relating to activities undertaken by ENERGEX at the request of customers or their agents (e.g. Retailers or contractors). The costs for these activities can be directly attributed to customers and service-specific charges can therefore be levied.
- **Quoted services** – are services for which the nature and scope cannot be known in advance irrespective of whether it is customer requested or there is an external event that triggers the need.

In the previous *regulatory control period* (under the Queensland Competition Authority (QCA)) fee based and quoted services were included under a group termed Excluded Distribution Services (EDS). At the start of a new regulatory year, the previous year's approved prices were taken as the base and were escalated by the All Groups CPI figure to produce prices for the next period.



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For the *2010-15 regulatory control period*, the prices for fee based services will be subject to an annual escalation process which will take into account a weighted average price increase for each of the components of the price formula.

Other material differences for the *2010-15 regulatory control period* include:

- overhead costs will be included as a \$ rate per direct spend and separated into on-costs and general overhead (previously overhead costs were considered as a \$ rate per hour);
- labour costs will reflect the mix of internal and external labour used to provide the service (previously only internal labour rates were used as the basis for estimates); and
- a methodology has been developed to attribute the cost of non-system assets used in the provision of services (previously QCA calculated and provided ENERGEX with the capital allowance figure).

### 7.2.2 Framework

*Alternative control services* are provided on an individual fee for service basis to retailers and end use customers. Depending on the service, ENERGEX will either provide the service for a fixed fee or quoted price.

The AER has proposed a formula based price cap form of control mechanism for fee-based services and quoted services over the *2010-15 regulatory control period*. This consists of:

- a schedule of fixed prices for fee-based services and a schedule of rates for quoted services for the first year of the *2010-15 regulatory control period*; and
- an annual escalation process for the remaining years of the *2010-15 regulatory control period*.

This formula outlined below for fee based services and quoted services has been designed to ensure prices will be representative of the efficient costs of providing and delivering the service.

From 1 July 2010, large customer connections will be classified as a quoted service under alternative control. Additional detail on the framework and approach for large customer connections is provided in Section 7.3.

### 7.2.3 Charging parameters

#### Regulatory Requirement

**Clause 6.18.2(b)(3) of the *Rules* states that:**

*A pricing proposal must set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.*



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In calculating the price for fee based and quoted services, ENERGEX will utilise the formula outlined below which includes cost parameters for different services to determine the price.

*Price = Labour + Contractor Services + Materials + Capital Allowance + GST*

- **Labour** (including overheads) – consists of all labour costs directly incurred in the provision of the service, labour on-costs, fleet on-costs and overheads. The labour cost for each service is dependent on the skill level, travel time, number of hours and crew size required to perform the service;
- **Contractor services** (including overheads) – reflects all costs associated with the use of external labour in the provision of the service, including overheads and any direct costs incurred as part of performing the service (e.g. traffic control, road closure permits);
- **Materials** (including overheads) – reflects the cost of materials directly incurred in the provision of the service, material storage and logistics on-costs and overheads;
- **Capital allowance** – represents a return on and return of capital for non-system assets used in the delivery of the service; and

#### 7.2.4 Tariffs

##### Regulatory Requirement

**Clause 6.18.2(b)(2) of the Rules states that:**

*A pricing proposal must set out the proposed tariffs for each tariff class.*

**The Final Determination from the AER requires:**

ENERGEX to provide, for fee based and quoted services, the proposed price for each service in the *regulatory year*, in addition to the volume of services provided and revenues collected from the provision of each service in the preceding *regulatory year*.

Prices for fee based services have been calculated using the above formula and forecast costs for labour, fleet on-costs, overheads and capital allowance.

The approved price schedule for fee based services in 2010-11 is provided in Appendix 2.2. The prices were determined using the AER's approved formula based price cap control mechanisms.

To meet statutory obligations and community expectations, it may be necessary to introduce a new fee based service during a regulatory year. The process which will be followed is outlined in Appendix 6.

The approved price schedule for quoted services (illustrative configuration) in 2010-11 is provided in Appendix 2.3. These prices were determined using the AER's approved formula based price cap control mechanisms. The AER's approved prices do not represent a binding capped price for an individual quoted service.



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ENERGEX has retained its current policy of not establishing a fixed price where variations in the precise nature of the services being sought mean that averaging would result in significant inequity for customers. The prices for quoted services will be calculated using the formula above to reflect the actual cost of service provision based on the specific requirements of the customer.

In line with the AER requirements, the volume of services and the revenues collected from the provision of each fee based and quoted service in the preceding regulatory year are provided in Appendix 2.4.

## 7.3 Large customer connections

### 7.3.1 Overview

In the previous *regulatory control period* the design and construction of large customer connection assets was classified as a prescribed service under the revenue cap. In the final framework and approach paper, the AER classified the design and construction of large customer connection assets to be an *alternative control service*.

The classification and control mechanism proposal for large customer connections departs from the current regulatory approach and is one of the key changes for ENERGEX and its customers in the *2010-15 regulatory control period*.

Accordingly from 1 July 2010, for new or upgraded connections (requested by a customer) of greater than 1 MVA or 4 GW.h pa (that is a tariff class of EG, CAC and ICC), the design and construction of the connection assets will be a quoted service and will be priced as detailed in Section 7.2.3.

As part of the new service classification, customers may choose to have the service provided by other suppliers subject to those suppliers complying with ENERGEX's Quality Assurance and materials specification requirements for the service.

The commissioning, operation and maintenance of all connection assets, including large connections, was determined by the AER to be a *standard control service*, which represents no departure from the current regulatory approach. In addition, the existing pricing methodology arrangements for the use of the shared network are unchanged.

### 7.3.2 Framework

The framework for new and existing large customer connections is provided in Figure 7.1 (classification of service) and Table 7.4 (pricing framework).

The design and construction of large customer connections will be classified as one of the following:

- **alternative control service** – for all new connections or upgrade to existing connections, which are paid for by the customer and gifted to ENERGEX. This includes an upfront payment for the design and construction of the connection asset, calculated in accordance with the formula in Section 7.2.3. These assets will form part of the



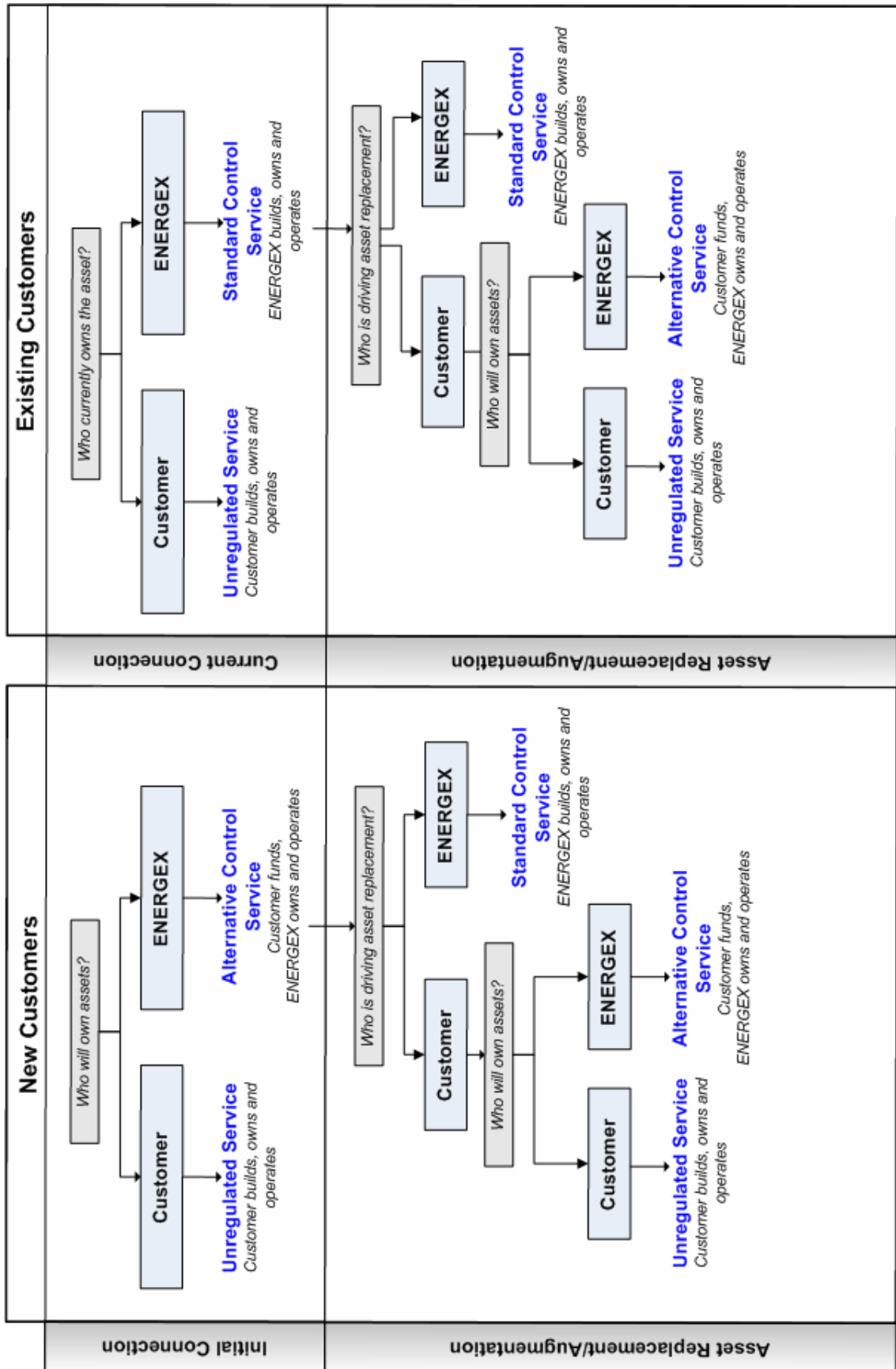
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Contributed Asset Base (CAB). Items in the CAB will have no return on capital or regulatory depreciation cost allocated to them. However there will be an allocation for operating and maintenance costs (O&M) recovered through DUOS as per the tariff cost allocation process.

- **standard control service** – applies to existing large customer connections assets which are owned and maintained by ENERGEX, or for ENERGEX driven asset replacement. These services will continue to form part of the Regulatory Asset Base (RAB). The methodology of calculating network charges for these customers is consistent with the previous *regulatory control period*. Accordingly, these connection assets have costs allocated to them for return on capital, regulatory depreciation and O&M as per the tariff cost allocation process detailed in Appendix 3.
- **unregulated service** – applies to connection assets which are funded, owned and operated by the customer. These services will attract no specific connection asset charges.



Figure 7.1 Service classification for large customer connections



**Table 7.4 Large customer connections pricing framework**

| Initial Connection Date                                              | Description                                              | Ownership <sup>1</sup> | Asset Classification           | Service Classification | Asset Base | Upfront Customer payment<br>(quoted price, relating to design and construction costs)                       | Tariff Charging Parameter<br>(Site Specific Fixed Charge)        |                                                     |
|----------------------------------------------------------------------|----------------------------------------------------------|------------------------|--------------------------------|------------------------|------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------|
|                                                                      |                                                          |                        |                                |                        |            |                                                                                                             | Recovery of depreciation and return on capital<br>(through DUOS) | Recovery of operating expenditure<br>(through DUOS) |
| Prior to 1 July 2010<br>(asset constructed under previous framework) | Existing connection                                      | ENERGEX                | Non-contributed                | SCS                    | RAB        | n/a                                                                                                         | ✓                                                                | ✓                                                   |
|                                                                      | Upgrade (ENERGEX driven) <sup>2</sup>                    | ENERGEX                | Non-contributed                | SCS                    | RAB        | n/a                                                                                                         | ✓                                                                | ✓                                                   |
|                                                                      | Upgrade (customer request)                               | ENERGEX (gifted)       | Contributed                    | ACS                    | CAB        | ✓                                                                                                           | n/a                                                              | ✓                                                   |
| Post 1 July 2010<br>(asset constructed under new framework)          | New Connection                                           | ENERGEX (gifted)       | Contributed                    | ACS                    | CAB        | ✓                                                                                                           | n/a                                                              | ✓                                                   |
|                                                                      | Upgrade (customer request)                               | ENERGEX (gifted)       | Contributed                    | ACS                    | CAB        | ✓                                                                                                           | n/a                                                              | ✓                                                   |
|                                                                      | Upgrade (ENERGEX driven) <sup>2</sup>                    | ENERGEX                | Non-contributed                | SCS                    | RAB        | n/a                                                                                                         | ✓                                                                | ✓                                                   |
| Asset constructed under either framework                             | Replacement (during warranty period for gifted assets)   | ENERGEX (gifted)       | Contributed                    | n/a                    | CAB        | n/a<br>(covered under warranty)                                                                             | n/a                                                              | ✓                                                   |
|                                                                      | Replacement (outside of manufacturers warranty period)   | ENERGEX (gifted)       | Contributed                    | ACS                    | CAB        | ✓                                                                                                           | n/a                                                              | ✓                                                   |
|                                                                      |                                                          | ENERGEX                | Non-contributed                | SCS                    | RAB        | n/a                                                                                                         | ✓                                                                | ✓                                                   |
|                                                                      | Relocation or rearrangement of assets (customer request) | ENERGEX                | Contributed or Non-contributed | ACS                    | n/a        | Remains as reimbursable works and will be included under the Alternative Control framework from 1 July 2010 |                                                                  |                                                     |
|                                                                      | Any service                                              | Customer               | n/a                            | Unregulated            | n/a        | No specific connection asset charges                                                                        |                                                                  |                                                     |

Standard Control Service (SCS), Alternative Control Service (ACS)

**Notes**

1. If the customer chooses to retain ownership of the asset, the service is unregulated and there are no specific connection asset charges.
2. An ENERGEX driven upgrade to a customer's connection assets could occur, when for network reasons, the connection arrangement needs to be altered.



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### 7.3.3 Charging parameters

ENERGEX will calculate the price for the design and construction of large customer connection assets in accordance with the formula outlined in Section 7.2.3.

### 7.3.4 Tariffs

#### Regulatory Requirement

**Clause 6.18.2(b)(2) of the *Rules* states that:**

*A pricing proposal must set out the proposed tariff for each tariff class.*

Due to the variability in the scope, the design and construction of large customer connections is a quoted service for which no specific prices are published. The price for the illustrative service configuration for large customer connections is shown in Appendix 2.3.

### 7.3.5 Transitional arrangements

Due to the change in service classification for the design and construction of large customer connections assets, the following transitional arrangements will be used:

1. the new *alternative control service* framework will apply to all new connection enquiries or applications received after 31 March 2010; and
2. for enquiries or applications received prior to and including 31 March 2010:
  - a. if there is a signed connection agreement or written acceptance of an offer (e.g. form 2226) in place by 30 June 2010, the existing (*standard control service*) framework will apply; or
  - b. if there is not a signed connection agreement in place at 30 June 2010, the new (*alternative control service*) framework will apply. If there is a concern that a specific project in this category should be treated differently, based on the status of discussions and negotiations, the project shall be referred to the Network Commercial Manager for advice and direction.



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## 8 Proposed tariff variations

### 8.1 Variations within the regulatory year 2010-11

SAC customer tariffs typically remain unchanged within the regulatory year. However, some SAC customers may participate in voluntary tariff trials during the regulatory year. These tariffs will be specifically designed to determine customer behaviour and may have special tariff elements for this purpose. More details on these tariffs are contained in Section 10.2.

The tariffs for larger customers are most likely to change where there are changes specific to an individual customer's connection arrangements and/or there is a material change in use of the site.

There is also the possibility of Government-mandated tariff changes. An example of such a change is the setting of prices for some of the *alternative control services* via Schedule 8 of the Queensland Electricity Regulation 2006. Another example is the solar feed-in tariff, where a new tariff was introduced with pricing mandated by Government.

Any tariff pricing changes for either *standard control services* or *alternative control services* during the 2010-11 regulatory year would be developed taking into account the same principles and objectives outlined in Section 4.1 of this proposal. Approval for new tariff prices would be sought from the AER via the process described in Appendix 6.

### 8.2 Variations between regulatory years 2009-10 and 2010-11

There are a number of changes to tariffs which have occurred relative to the tariffs approved for the 2009-10 year, including:

- the introduction of a power factor adjustment (kV.A based) which is used in the calculation of demand and capacity charges for ICCs and CACs (see Appendix 1.3);
- the separation of existing street lighting services into a *standard control service* for the conveyance of electricity to street lights and an *alternative control service* for the provision, construction and maintenance of street lights;
- a limited building block approach to determine the efficient costs of providing street lighting services under the price cap control mechanism in the first year of the *regulatory control period* and establishment of a price path for the remaining period;
- the application of a price cap control mechanism to ENERGEX's *alternative control services* for the *2010-15 regulatory control period*;
- a change to the method of calculating price increases for fee based services from a general CPI increase to an average weighted price increase based on the individual components of the price formula; and
- the classification of the design and construction of large connection assets as an *alternative control service*, included in the quoted services group.



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## 9 Transmission cost recovery

### Regulatory Requirement

#### Clause 6.18.2(b)(6) of the *Rules* states that:

A *pricing proposal* must set out how charges incurred by the *Distribution Network Service Provider* for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous *regulatory year*.

#### Clause 6.18.7 of the *Rules* states that:

- (a) A *pricing proposal* must provide for tariffs designed to pass on to customers the charges to be incurred by the *Distribution Network Service Provider* for transmission use of system services.
- (b) The amount to be passed on to customers for a particular *regulatory year* must not exceed the estimated amount of the transmission use of system charges for the relevant *regulatory year* adjusted for over or under recovery in the previous *regulatory year*.
- (c) The extent of the over or under recovery is the difference between:
  - (1) the amount actually paid by the *Distribution Network Service Provider* by way of transmission use of system charges in the previous *regulatory year*; and
  - (2) the amount passed on to customers by way of transmission use of system charges by the *Distribution Network Service Provider* in the previous *regulatory year*.

#### The Final Determination from the AER requires:

ENERGEX, in accordance with clause 6.12.1(19) of the *Rules*, to maintain a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with Appendix E of the Final Determination.

In accordance with Clauses 6.18.2(b)(6) and 6.18.7 of the *Rules*, tariffs outlined in this initial pricing proposal will allow for the pass through of charges for TUOS services, including any adjustments for under or over recovery. To comply with the *Rules*, information reported as part of this initial pricing proposal includes:

#### Payments:

- regulated transmission charges paid to Transmission Network Service Providers (TNSPs);
- avoided TUOS payments to embedded generators; and
- payments made to other DNSPs for use of their network.



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**Receipts:**

- payments received from network users; and
- payments received from other DNSPs.

**Adjustments for under/over recovery:**

- difference between receipts and payments.

## **9.1 Payments**

### **9.1.1 Transmission charges paid to TNSPs**

ENERGEX connects to the Powerlink network at multiple connection points. Powerlink, as a regulated TNSP, recovers its revenue from directly connected customers and DNSPs connected to its network.

In accordance with the connection agreement with Powerlink, ENERGEX is required to pay TUOS charges to Powerlink on a monthly basis.

### **9.1.2 Avoided TUOS for embedded generators**

Clause 5.5(h) of the NER requires ENERGEX to remit avoided TUOS usage charges (avoided TUOS) to EGs who have generated electricity and transmitted this energy into ENERGEX's network.

In accordance with the *Rules*, where prices for the locational component of prescribed TUOS services were in force at the relevant Transmission Network Connection Point (TNCP) throughout the relevant financial year, ENERGEX shall (for each EG):

- (a) determine the charges for the locational component of prescribed TUOS services that would have been payable by ENERGEX had the EG not injected any energy at its connection point during that financial year;
- (b) determine the amount by which the charges calculated in (a) exceed the amount for the locational component of prescribed TUOS services actually payable by ENERGEX; and
- (c) credit the value from (b) to the EG account.

In 2010-11, avoided TUOS payments will be remitted in the form of a lump sum payment after 30 June 2011, in line with previous years.

Avoided TUOS payments to EGs by ENERGEX reflect the avoided costs of upstream transmission network reinforcement to South East Queensland. As such, the benefits primarily relate to all customers – that is, avoided TUOS does not solely impact on the TNCP to which the EG is connected. Avoided TUOS has therefore been assigned to all tariff classes.



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### **9.1.3 Payments to other DNSPs**

In contingency circumstances, Country Energy provides supply from its Terranora Substation in northern NSW to ENERGEX's Kirra Zone Substation. Under this arrangement, Country Energy requires ENERGEX to pay for the use of its assets.

The charges established by Country Energy in respect of this arrangement are based on approved rates for each month in which the alternate supply is utilised. Based on these charges, it is expected the total estimated charges for Country Energy in 2010-11 will amount to \$360,000. These costs have been incorporated into the TUOS costs for the Mudgeeraba TNCP and therefore result in the charges from the use of Country Energy network being passed through to users.

## **9.2 Receipts**

### **9.2.1 Recovery of TUOS through tariffs**

A description of how TUOS is recovered through standard control tariffs is given in Section 6.2. The total revenue received is indicated in Table 9.1.

### **9.2.2 Payments received from other DNSPs**

ENERGEX does not currently receive any TUOS payments from other DNSPs.

## **9.3 TUOS unders and overs account**

As part of the requirements of the Final Determination, the AER requires ENERGEX to provide amounts for the following entries in its TUOS unders and overs account for the most recently completed regulatory year (t-2) and the next regulatory year (t):

1. The opening balance for each year. The opening balance for year t-2 should be zero.
2. The amount of revenue recovered from TUOS charges applied in respect of that year, less the amounts of all transmission related payments made by ENERGEX in respect of that year.
3. An interest charge for two years related to the net amount in item 2 for year t-2. This adjustment should be calculated using the approved nominal weighted average cost of capital (WACC). No such adjustment applies to the net amount in item 2 for year t as no such adjustment was required by the QCA.
4. The total of items 1 to 3 to derive the closing balance for each year.

Table 9.1 outlines the TUOS unders and overs calculation and provides separate identification of any under or over recovery relating to prior years included in the current year revenue. This distinction allows correct calculation of the under / over recovery of the respective year.



**Table 9.1 Calculation of TUOS unders and overs account**

|                                                     | <b>2008-09<br/>actual<br/>(\$'000)</b> | <b>2010-11<br/>forecast<br/>(\$'000)</b> |
|-----------------------------------------------------|----------------------------------------|------------------------------------------|
| Revenue from TUOS charges                           | 260,329                                | 336,685                                  |
| Less under/over adjustment approved for year t-2    | 6,430                                  | n/a                                      |
| Total transmission related payments                 | 246,683                                | 345,180                                  |
| Transmission charges to be paid to TNSPs            | 246,233                                | 344,320                                  |
| Avoided TUOS payments                               | 450                                    | 500                                      |
| Inter – DNSP payments                               | -                                      | 360                                      |
| <b>Under/over recovery for regulatory year</b>      | <b>7,216</b>                           | <b>-8,495</b>                            |
| <b>TUOS Unders &amp; Overs Account</b>              |                                        |                                          |
| Nominal WACC                                        | 8.5%                                   | n/a                                      |
| Opening Balance                                     | -                                      | 8,495                                    |
| Under/over recovery for regulatory year             | 7,216                                  | 8,495                                    |
| Interest on under/over recovery for regulatory year | 1,279                                  | n/a                                      |
| <b>Closing Balance</b>                              | <b>8,495</b>                           | <b>0</b>                                 |

The AER requires that the amounts used in the table above for the most recently completed regulatory year (t-2) are audited.

ENERGEX believes this requirement has been met as the figures are either:

- (a) audited as part of the regulatory accounts audit undertaken by Queensland Audit Office (QAO);
- (b) provided by the AER in the Final Determination; or
- (c) calculated based on underlying figures which are included under points (a) and/or (b) above.

Amounts for the next regulatory year (t) are forecast amounts.

In proposing variations to the amount and structure of TUOS charges for a given regulatory year t, ENERGEX will achieve a zero expected balance on its TUOS unders and overs account at the end of each regulatory year in the *regulatory control period*.



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# 10 Tariff development

ENERGEX undertakes a continual program to improve the structure of its tariffs. Each year ENERGEX publishes a discussion paper to inform stakeholders of the tariff structures that are under consideration within the medium term. The current paper can be found on the ENERGEX website and a summary of the proposed changes to the structure of current tariffs or potential new tariffs to be introduced is provided below.

## 10.1 Structural changes to tariffs

### 10.1.1 kV.A tariffs for large customers

#### 10.1.1.1 Rationale

From ENERGEX's perspective, providing the correct pricing signals will assist in ensuring that the utilisation of its network is at its most efficient – an outcome which is of benefit to all customers, as it will reduce the need to spend additional funds on increasing the capacity of the network, thereby minimising tariff increases.

The two methods of measuring utilisation of the network, or customer demand, are kW or kV.A. The power factor, or kV.A, is important because transmission and distribution systems must be designed and built to manage the need for real power as well as its associated reactive component – the total power. The total power can be greater than 50 per cent more than real power alone if the power factor is poor. Ideally, power factor should be as close to unity as possible. Therefore the further it moves away from one, the more network capacity is needed to supply the same kW demand. This is demonstrated in Table 10.1 below (where total power (kV.A) = demand (kW) / power factor).

**Table 10.1 Impact of power factor on the network**

|            | <b>Demand<br/>(kW)</b> | <b>Power<br/>Factor</b> | <b>Total Power<br/>(kV.A)</b> |
|------------|------------------------|-------------------------|-------------------------------|
| Customer 1 | 1,600                  | 0.7                     | 2,286                         |
| Customer 2 | 1,600                  | 0.8                     | 2,000                         |
| Customer 3 | 1,600                  | 0.9                     | 1,778                         |
| Customer 4 | 1,600                  | 1.0                     | 1,600                         |

A tariff based on kV.A would therefore be a more accurate measure of a customer's impact on the network, relative to a tariff based on kW, as it better reflects the costs imposed on the network by users. Accordingly, a kV.A tariff would better meet ENERGEX's pricing principles as it is more cost-reflective and will encourage more efficient use of the network.



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#### 10.1.1.2 Consultation process

ENERGEX understands the importance of a robust and consultative approach to new electricity pricing and tariff structures. ENERGEX has undertaken an extensive consultation process to incorporate the development of kV.A tariffs.

The following summarises the consultation process which has been undertaken from 2006 to 2010:

##### **Discussion Papers**

ENERGEX has published a number of discussion papers on its website to provide information to customers on proposed changes to tariffs. ENERGEX welcomed feedback from all stakeholders, including customers, retailers, policy makers and Regulators, on these papers. Feedback received from all stakeholders throughout the consultation process was considered by ENERGEX in the development of the final position paper.

- Development of Network Tariff Structures Discussion Paper, December 2006
- Development of Network Tariff Structures Discussion Paper, December 2007
- Development of kV.A Network Tariffs Consultation Paper, August 2008
- Development of kV.A Network Tariffs Preliminary Position Paper, October 2008
- Development of kV.A Network Tariffs Final Position Paper, December 2008

##### **Customer forums**

Customer forums were held in September 2007, November 2007, June 2008, September 2008 and June 2009. These customer forums were well attended by stakeholders. More than 50 customers, retailers and industry organisations attended the September 2008 forum, which focused specifically on kV.A tariffs.

In addition, ENERGEX was invited to present on the issue of kV.A tariffs at the Energy Users Association of Australia (EUAA) members meeting in November 2008.

##### **Paper trials**

In September 2008, all large customers (ICCs and CACs) were sent a letter comparing their network charges based on both a kW tariff and a kV.A tariff with the exception of a small number of customers where there was insufficient data. Actual customer data was used to illustrate how their network charges would have varied if kV.A tariffs had been applied to their 2007-08 consumption. Each letter also provided an estimate of the kV.A impact in 2008-09 based on forecast consumption data. The paper trial was conducted using a power factor neutral point of 0.80.

In November 2008, following the publication of the preliminary position paper, a second paper trial utilised the same consumption data as was used in September 2008 but with the power factor neutral point of 0.85 for CACs and 0.90 for ICCs, as proposed in the preliminary position paper. The power factor adjustment was applied to adjust the demand charge and capacity charge elements, reflecting the change in position since the August 2008 paper.



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In 2008 ENERGEX also published a tool on its website to enable demand metered SACs to calculate billing information for themselves by entering their own demand, energy and kV.A information. While ENERGEX isn't proposing to introduce kV.A tariffs to these customers at this time, the purpose of the calculator was to enable demand metered SACs to understand the likely impact of kV.A tariffs should ENERGEX propose to implement this type of tariff structure in the future.

In December 2009, a third paper trial was conducted which provided customers with an estimate of the likely impact of a kV.A tariff based on the current 2009-10 tariffs. The prices were applied to meter data that was available to ENERGEX at the time the calculation was performed plus forecast consumption and demand for the remainder of the 2009-10 year. The December 2009 paper trial used the power factor neutral point of 0.85 for CACs and 0.90 for ICCs, as outlined in the final position paper.

### **Bilateral consultation**

In addition to the more formal communication forums, ENERGEX has also actively undertaken bilateral consultation. ENERGEX has held discussions with more than 60 customers since 2008 to discuss kV.A tariffs, both customer initiated and ENERGEX initiated, by telephone, email or face-to-face. Meetings were also held during 2007 and 2008 with the Department of Mines and Energy (DME), the QCA and the EUAA.

In addition, ENERGEX contacted Queensland retailers via email to inform them of developments in the kV.A tariff consultation process, in August 2008 and November 2008. A number of retailers also attended the customer forum held in September 2008. In December 2009, retailers were contacted via email to advise them that ENERGEX will be including 2 new Tariff Component Codes for the planned implementation on 1 July 2010 of a kV.A tariff. ENERGEX advised that it will update the Queensland B2B Process Specification - Network Billing Section 3.10.3.1 and that these new codes will only apply to ICC and CAC NMIs. Retailers were also provided with an example letter which was sent to the customers as part of the 3rd paper trial and a copy of the kV.A calculation algorithm to help them better understand the methodology and impact. Using this methodology the retailers were able to implement the changes required to current billing systems and procedures.

#### **10.1.1.3 Establishment of the kV.A tariff**

Following proactive engagement with stakeholders on the issue of kV.A, ENERGEX proposed to introduce kV.A tariffs to the tariff structure as it better meets its pricing principles. ENERGEX's 2009-10 Pricing Principles Statement (approved by the QCA) signalled the introduction of this change.

ENERGEX will be introducing kV.A tariffs for ICC and CAC customers from 1 July 2010. Network charges for ICC and CAC customers will be calculated initially using the existing kW based approach. The charges will then be adjusted up or down by a relative amount based on the difference between the customer's measured power factor and the power factor neutral point. The adjustment will be applied to the demand and capacity charge elements. This methodology is described further in Appendix 1.3. An update of the billing systems for both ENERGEX and the retailers has commenced to reflect the new kV.A calculation



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methodology. The kV.A methodology will be revenue neutral for the ICC tariff class and each of the CAC tariff classes.

ENERGEX intends to explore the opportunity to move to a straight kV.A tariff<sup>3</sup>, reflective of current practices in other Australian jurisdictions, over the medium term. The power factor adjustment methodology outlined above is a transitional method to help customers understand the concept of kV.A and the impact of kV.A on the network and network charges. Further, the proposed method has a limited life as customers change their behaviour.

## 10.2 Tariff trials

ENERGEX is implementing tariff trials in a number of areas. These trials provide ENERGEX with information on potential customer behaviour with respect to new tariffs.

### 10.2.1 Controlled load tariffs

Controlled load is load that can be managed at ENERGEX's discretion within specific periods as agreed with the customer. Controlled load provides a means to manage peak demand during times of maximum network utilisation. This reduces the capacity requirements on the network. Load may be interrupted without loss of utility benefits to customers or, where customers volunteer, some utility benefit may be forgone for a financial and/or environmental reward.

#### 10.2.1.1 Air-conditioning direct load control

Domestic and commercial air-conditioning load continues to be a significant contributor to peak demand. During the *2010-15 regulatory control period*, ENERGEX will build on the early success of its 'time for a cool change' (Cool Change) trials, by rolling out the technique of air-conditioning compressor 'cycling' more widely across ENERGEX's distribution area.

The Cool Change trials, undertaken in the northern suburbs of Brisbane from December 2007 and continuing to 2011, involve more than 2000 residential volunteers.

#### 10.2.1.2 Pool pump direct load control

The Cool Change program has recently been expanded to include trials with pool pumps. Estimates indicate pool pumps contribute significantly to system peak demand and initial analysis has indicated that benefits exist in shifting pool pump usage from peak use times. Accordingly, in addition to offering and promoting a better rate for electricity used by pool pumps at non-peak times through tariff 33, ENERGEX will be conducting further trials in 2010 to leverage off the learnings of the Cool Change air-conditioning trials. This pool pump trial will utilise a new generation of audio frequency load control technology similar to that employed in the Cool Change air-conditioning trials. More than 500 households have already subscribed.

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<sup>3</sup> Initially a kV.A adjustment factor will be applied to the demand and capacity charge elements (currently a \$/kW charge). In the future, a straight kV.A tariff with an associated charging parameter in \$/kV.A may be used.



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### 10.2.1.3 Hot water load control

In regard to hot water load control, ENERGEX proposes to conduct a campaign across SEQ to offer incentives to convert from a continuous supply tariff (retail notified tariff 11) to a less expensive off-peak tariff that provides supply for a minimum eight hours (controlled load 1 – retail tariff 31) or 18 hours (controlled load 2 – retail tariff 33) per day. In addition, ENERGEX is reviewing switching times with the view of optimising the amount of load managed during summer peaks.

## 10.2.2 Reward-based tariff trials

ENERGEX intends to examine the potential benefits of reward-based tariffs as a method of encouraging consumers to reduce or limit demand at peak times. The reward-based tariff operates by providing an incentive payment to customers for load reductions in certain periods. Two such tariffs that are being investigated are dynamic peak pricing tariffs and Time of Use (ToU) tariffs.

This project will conduct pricing trials in SEQ and identify the benefits from ToU and dynamic pricing tariffs. In addition to research and analysis on load limiter technology (a device used to limit the maximum demand per household), these trials will provide an understanding of customer acceptance and behaviour towards tariffs of this nature and develop ENERGEX's understanding of the effectiveness of such tariffs. These learnings will feed into future tariff policy development.

The trials and research are a precursor to the potential deployment of reward-based tariffs amongst residential customers in SEQ.

## 10.2.3 Residential Time of Use (ToU) Tariff

The objective of ToU tariffs is to provide a signal to customers to encourage a transfer of usage from periods of high demand to periods of low demand. It does so by offering a lower tariff during off-peak periods and higher tariffs during peak periods. ToU pricing encourages more efficient use of the network as customers switch nonessential electricity use to off-peak periods, reducing peak demand and ultimately benefiting all customers through reduced network costs over the long term. ToU pricing also allows customers the flexibility to optimise their own electricity costs.

ToU tariffs were introduced for small and medium non-demand business customers from 1 July 2007 and for large business customers from 1 July 2008. ToU tariffs for business customers are based on a simple peak/off-peak differential.

ENERGEX currently has around 200,000 small customers with multi-function electronic meters that are capable of supporting ToU tariffs. ENERGEX intends to continue investigating the feasibility of introducing a ToU network tariff, potentially starting with these customers. However, ToU tariffs would only be considered for introduction by ENERGEX after an extensive consultation process.

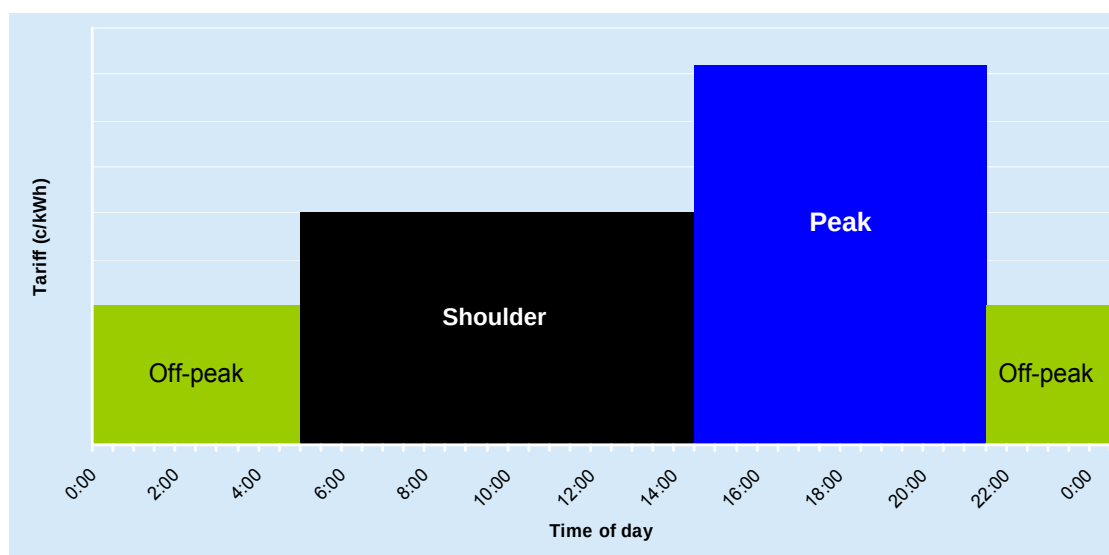
Another factor that could drive the creation of a ToU network tariff is the outcome from the QCA's retail tariff review. This review recommended the creation of a voluntary ToU retail



tariff for residential customers with an interval meter. If this tariff is introduced then ENERGEX could be compelled to create a complementary ToU network tariff.

Residential ToU tariffs may be structured with differential tariffs for peak, shoulder and off-peak periods, as illustrated in the following diagram. Further, there may be different tariffs depending on the season (e.g. higher tariffs in summer).

**Figure 10.1 Example Time of Use tariff using a shoulder, peak and off-peak rate**



ENERGEX is exploring the opportunity to reconstruct the current ToU tariff to incorporate both a shoulder period and/or the introduction of a seasonal element. This initiative is designed to better reflect the expected patterns of utilisation and enhance the signal to customers of their impact on the network.

### 10.3 Proposed removal of tariffs

ENERGEX does not anticipate the removal of any tariffs over the 2010-11 regulatory year.

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# 11 Compliance with regulatory requirements

## 11.1 DUOS unders and overs account

### Regulatory Requirement

#### **The Final Determination from the AER requires:**

ENERGEX to keep and provide information on a distribution use of system (DUOS) unders and overs account under clause 6.18.2(b)(7) of the *Rules*.

As part of the requirements of the Final Determination, the AER requires ENERGEX to provide entries in its DUOS unders and overs account for the most recently completed regulatory year (t-2) and the next regulatory year (t).

Where ENERGEX has an under or over recovery greater than two percent, the adjustment to revenue may span two or more years as allowed for in the tolerance limits set out in Section 4 of the Final Determination. Where the under/over recovery is not cleared in one year, additional interest is charged on the remaining balance to index it to the year in which it is adjusted.

Table 11.1 outlines the DUOS unders and overs calculation and provides separate identification of any prior period under or over recovery included in the current year revenue. This distinction allows correct calculation of the under / over recovery for the respective years.



**Table 11.1 Calculation of DUOS unders and overs account**

|                                                        | 2008-09<br>actual<br>(\$'000) | 2010-11<br>forecast<br>(\$'000) |
|--------------------------------------------------------|-------------------------------|---------------------------------|
| Revenue from DUOS charges                              | 910,553                       | 1,084,467                       |
| Less under/over adjustment approved for year t-2       | -5,886                        | n/a                             |
| Less MAR for the relevant year                         | 874,986                       | 1,108,867                       |
| Allowed revenues ( $AR_t$ )                            | 874,986                       | 1,135,060                       |
| Service quality performance reward/penalty ( $S_t$ )   | -                             | -                               |
| Capital contribution overs/unders adjustment ( $C_t$ ) | -                             | 902                             |
| Transitional adjustments ( $Transitional_t$ )          | -                             | -27,095                         |
| Approved pass throughs ( $Passthrough_t$ )             | -                             | -                               |
| <b>Under/over recovery for regulatory year</b>         | <b>41,453</b>                 | <b>-24,400</b>                  |
| <b>DUOS unders and overs account</b>                   |                               |                                 |
| Nominal WACC                                           | 8.5%                          | n/a                             |
| Opening balance                                        | -                             | 48,800                          |
| Interest on opening balance                            | -                             | n/a                             |
| Under/over recovery for regulatory year                | 41,453                        | -24,400                         |
| Interest on under/over recovery for regulatory year    | 7,347                         | n/a                             |
| <b>Closing Balance</b>                                 | <b>48,800</b>                 | <b>24,400</b>                   |

The AER requires that the amounts used in the table above for the most recently completed regulatory year (t-2) are audited.

ENERGEX believes this requirement has been met as the figures are either:

- (a) audited as part of the regulatory accounts audit undertaken by Queensland Audit Office (QAO);
- (b) provided by the AER in the Final Determination; or
- (c) calculated based on underlying figures which are included under points (a) and/or (b) above.

Amounts for the next regulatory year (t) are forecast amounts.

In proposing variations to the amount and structure of DUOS charges, ENERGEX will achieve an expected zero balance on its DUOS unders and overs accounts at the end of each regulatory year in the next *regulatory control period*, unless ENERGEX can demonstrate for a given year that such an adjustment exceeds the agreed tolerance limits set out in Section 4 of the Final Determination.



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The proposed prices for year t will be based on the sum of the MAR for year t plus any under/over recovery for year t, up to 2 per cent of ENERGEX's revenues unless otherwise agreed between the AER and ENERGEX.

## 11.2 Compliance with avoidable and stand-alone cost requirements

### Regulatory Requirement

Clause 6.18.5 of the *Rules* states that:

- (a) For each *tariff class*, the revenue expected to be recovered should lie on or between:
- (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and
  - (2) a lower bound representing the avoidable cost of not serving those customers.

### 11.2.1 Standard control services

In line with the NER, the revenue expected to be recovered from each tariff class will lie on or between the bounds of stand alone and avoidable costs. This requirement ensures that the revenue recovered from each tariff class is cost reflective and free from cross subsidies.

Section 11.2.1.2 describes how this requirement is met, while Table 11.3 provides estimates which indicate that the expected revenue to be recovered from tariffs for 2010-11 is between the two bounds.

#### 11.2.1.1 Definitions

Stand-alone costs are the costs that would be incurred to service a group of customers (referred to as a tariff class) on a stand-alone basis.

- Stand alone costs can be estimated in two ways. Firstly, from first principles the cost can be estimated by assuming that a new hypothetical network is constructed solely for the particular tariff class. Secondly, the costs can be estimated by calculating the cost of using the existing network to supply the particular tariff class on a stand alone basis. For the purposes of this document, ENERGEX will use the second method. It is noted that due to economies of scale, the total stand alone costs based on the first method will be greater than the figure calculated using this second method. However, as this is the basis upon which the expected revenue (and subsequent tariffs) is calculated and the estimated stand alone cost represents the upper bound, this is considered to be a satisfactory method for this purpose.
- Avoidable costs reflect the costs that would be avoided if a particular group of customers (referred to as a tariff class) were no longer connected to the network, assuming that all other customers remain connected.



### 11.2.1.2 Background

ENERGEX calculates tariffs using a Distribution Cost of Supply (DCOS) model. The DCOS model generates DUOS tariffs based on the full distribution of the building block costs approved by the AER. Appendix 3 outlines the tariff cost allocation process which is used by ENERGEX.

The DCOS model has also been used to estimate the stand alone and avoidable costs for each tariff class. Table 11.2 outlines which of the DCOS cost categories are used in the calculation of the two price boundaries.

In addition, the conversion of the DCOS cost categories to the tariff charging parameters is provided in Table 11.2.

**Table 11.2 DCOS categories used in price boundary calculations and the conversion of DCOS categories into tariffs**

| DCOS Cost Category              |                                            | Pricing Boundaries |                  | Tariff Charging Parameter                        |                                     |                                  |
|---------------------------------|--------------------------------------------|--------------------|------------------|--------------------------------------------------|-------------------------------------|----------------------------------|
|                                 |                                            | Avoidable cost     | Stand alone cost | Fixed or Service Availability Charge (\$/period) | Volume Charge <sup>2</sup> (c/kW.h) | Capacity / Demand Charge (\$/kW) |
| Network O&M                     | Non-contributed Connection Assets          | ✓                  | ✓                | ✓                                                |                                     |                                  |
|                                 | Contributed Connection Assets              | ✓                  | ✓                | ✓                                                |                                     |                                  |
|                                 | Network Assets                             |                    | ✓                |                                                  | ✓ <sup>1</sup>                      | ✓                                |
| Network Regulatory Depreciation | Non-contributed Connection Assets          | ✓                  | ✓                | ✓                                                |                                     |                                  |
|                                 | Contributed Connection Assets <sup>3</sup> |                    |                  |                                                  |                                     |                                  |
|                                 | Network Assets                             |                    | ✓                |                                                  | ✓ <sup>1</sup>                      | ✓                                |
| Network Return on Capital       | Non-contributed Connection Assets          | ✓                  | ✓                | ✓                                                |                                     |                                  |
|                                 | Contributed Connection Assets <sup>3</sup> |                    |                  |                                                  |                                     |                                  |
|                                 | Network Assets                             |                    | ✓                |                                                  | ✓ <sup>1</sup>                      | ✓                                |
| Common Services                 |                                            |                    | ✓                |                                                  | ✓                                   |                                  |
| Non-System                      |                                            |                    | ✓                | ✓                                                | ✓                                   | ✓                                |

1 Applicable to customers on volume based network tariffs.

2 The structure of the volume charge can incorporate either a flat rate or a Time of Use volume charge.

3 There is no regulatory depreciation or return on capital for contributed connection assets.



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### **Lower bound test (avoidable cost)**

As shown in Table 11.2, the avoidable costs for a tariff class include the cost of connection and the costs of operating and maintaining that connection. The fixed or standing charge for the customer includes these costs, making it the floor price. Therefore, any use of the shared network will incur additional charges (in the form of the volume and/or capacity/demand charge parameters), taking the charge paid by any customer above the avoidable cost of supply (the economic cost floor).

### **Upper bound test (stand alone cost)**

In the DCOS model the infrastructure, operating and maintenance costs for upstream assets are shared across multiple customers and tariff classes. For this reason, the allocated cost of supply for each tariff class will be equal to, or below the stand-alone costs of supply.

In the case of smaller network customers connected at the distribution level (generally 11kV and below), the allocated cost will be well below stand-alone costs as the costs for high voltage assets are shared with larger customers.

For larger network customers connected at the sub-transmission level (33kV and above), the allocated cost model includes a site-specific parameter of supply network costs. The cost-based tariffs for CACs take into account the specific connection costs as well as an allocation of the upstream shared network costs. ICC cost-based tariffs are determined by mapping the actual supply network and allocating the relevant proportion of costs to the customers on the basis of their use of that network.

Therefore as there is an allocation of costs, and/or the full network costs are allocated in the case of a single-user asset, the revenue recovered from tariffs must be equal to or below the stand-alone cost of supply (economic cost ceiling).

#### **11.2.1.3 Estimates**

The methodology used by ENERGEX in calculating stand alone and avoidable costs is described below.

The avoidable cost estimate for each tariff class has been developed assuming:

- Avoidable capital - the annual capital charge is taken as the return on capital and depreciation allocations for the non-contributed connection assets required to service the tariff class.
- Avoidable O&M - the annual O&M costs are taken as those costs allocated to all connection assets for each tariff class in the DCOS model. It is assumed that the shared O&M costs for common and non-system assets are not avoidable as they are incurred irrespective of whether one particular tariff class is no longer connected.
- Total avoidable costs - are a sum of the above and are represented as an annual charge in Table 11.3 below.



The stand alone cost estimate for each tariff class has been developed assuming:

- The existing network is used to supply the particular tariff class on a stand alone basis.
- Stand alone capital – the annual capital charge is taken as the return on capital and depreciation allocations for the non-contributed connection assets and all network (shared) assets required to service the tariff class.
- Stand alone O&M - the annual O&M costs are taken as the total allocated for each tariff class in the DCOS model. This includes costs for connection and network assets, common services and non-system assets.
- Total stand alone costs - are a sum of the annual capital and O&M charges above and are represented as an annual charge in Table 11.3 below.

**Table 11.3 Stand alone and avoidable cost boundaries for 2010-11**

| <b>Tariff Class</b> | <b>Avoidable cost (\$m)</b> | <b>Expected revenue (\$m)<br/>(excl. side constraint adjustment)</b> | <b>Stand alone cost (\$m)</b> |
|---------------------|-----------------------------|----------------------------------------------------------------------|-------------------------------|
| ICC                 | 9.1                         | 25.2                                                                 | 28.4                          |
| CAC - 33kV          | 1.4                         | 2.9                                                                  | 3.1                           |
| CAC - 11kV Bus      | 3.9                         | 12.6                                                                 | 14.2                          |
| CAC - 11kV Line     | 7.1                         | 62.3                                                                 | 75.3                          |
| EG                  | 0.6                         | 1.3                                                                  | 1.8                           |
| SAC Demand          | 17.2                        | 164.5                                                                | 189.0                         |
| SAC Non-demand      | 118.1                       | 815.7                                                                | 1,042.5                       |

### 11.2.2 Alternative control services

The formula used in the calculation of prices for fee based services and quoted services has been designed to ensure prices will be representative of the efficient costs of providing and delivering the service and signal the economic costs of service provision by being subsidy free. Prices based on this formula will be cost reflective, representing costs derived through the same allocative mechanism as that used to determine costs for *standard control services*, in accordance with the AER approved Cost Allocation Method (CAM). The prices for each tariff class within *alternative control services* will be between the bounds of avoidable and stand alone costs due to the economies of scale in providing each service.

The avoidable cost for a particular service is equivalent to the direct labour and materials cost. Overhead costs will be incurred regardless of whether the service is provided or not. Consequently, the price calculated using the formula in Section 7.2.3 (which includes an allowance for overhead costs) will be greater than the avoidable cost.



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The stand alone cost is equal to the costs of serving each tariff class within *alternative control services* on a stand alone basis. For example, the stand alone cost would require the use of dedicated resources and assets (including costs associated with corporate support, capital allowance for non-system assets, etc.). As these costs can be shared among all of the tariff classes within *alternative control services*, the cost calculated for each individual service will be less than the stand alone cost.

## 11.3 Long run marginal cost

### Regulatory Requirement

#### Clause 6.18.5(b)(1) of the Rules states that:

A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:

- (1) must take into account the long run marginal cost for the service or, in the case of a *charging parameter*, for the element of the service to which the *charging parameter* relates.

### 11.3.1 Standard control services

Marginal cost is the change in total cost that arises when the quantity produced changes by one unit. In the case of an electricity network, the marginal cost is the cost incurred from one additional customer connecting to the network.

It is important that prices for access to the electricity network are set at a minimum of marginal cost, to ensure a new customer is paying at least the direct costs of their connection to the network.

Marginal costs can be calculated as either short run marginal costs (SRMC) or long run marginal costs (LRMC). In the short run investment in capacity is fixed and therefore the SRMC refers to the cost of a customer connecting to the network but using only the existing capacity in the network. In the long run, investment in capacity is variable and hence LRMC indicates an estimate of the cost of connecting the customer when an augmentation to the capacity of the network is necessary. Pricing on the basis of LRMC therefore assumes that prices should be based on the cost of meeting an increase in demand over an extended period of time.

In the electricity market, as demand on the network increases, an expansion of network capacity is required to accommodate the additional demand. In order to provide correct signals for customers, prices must take into account these additional capital costs. The LRMC should be considered in setting tariffs as it can provide appropriate signalling to customers and will assist in achieving a stable tariff structure over the forecast period.



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One method which is used to estimate the LRM is the Average Incremental Cost (AIC) method:

$$\text{LRMC (AIC)} = \frac{\text{PV(Capex)} + \text{PV(Opex)}}{\text{PV(Incremental Demand)}}$$

where, *PV (Capex) and PV (Opex) represent the costs associated with meeting the additional demand over the regulatory control period.*

This calculation method gives rise to an estimate which smoothes out irregular capital expenditure over time, whilst providing an indication of the costs associated with the increased demand. Thus, the LRM indicates the level at which future increments of output must be priced to ensure total cost recovery given forecast demand.

The incremental capital and operating costs associated with increased demand are included in the ENERGEX forecast capital and operating expenditure programs for 2010-15. ENERGEX receives a return on these (and other) costs in the form of a return on capital and depreciation charges as approved by the AER. These costs are then allocated to each tariff class as part of the DCOS model.

ENERGEX designs tariffs to contain a combination of charging parameters in order to reflect LRM and recover the total allowable revenue:

- where appropriate, a capacity charge may take into account the long term demand peak and can provide effective pricing signals to customers of the next increment of load;
- a demand charge is used to take into account short term peaks in demand;
- energy charges are used where appropriate due to limitations with current metering; and
- fixed charges are used to ensure the remaining costs including the costs associated with connection assets are recovered.

### **11.3.2 Alternative control services**

Prices for *alternative control services* are calculated based on the current costs to provide a particular service. The LRM is taken into consideration when determining prices for *alternative control services*. However, given that there is only a single charge parameter for each *alternative control service*, prices have been set at average cost. This ensures total costs are recovered for these services, as allowed for under clause 6.18.5(c).



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## 11.4 Transaction costs

### Regulatory Requirement

**Clause 6.18.5(b)(2)(i) of the *Rules* states that:**

A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:

- (2) must be determined having regard to:
  - (i) transaction costs associated with the tariff or each *charging parameter*.

For each tariff, ENERGEX has selected a number of charging parameters (identified in Section 6.2 for *standard control services* and Section 7.1.3 and 7.2.3 for *alternative control services*). Each combination of charging parameters in a tariff has been selected to adequately reflect the need for both fixed and variable components.

ENERGEX has sought to simplify pricing structures without sacrificing economic functionality, whilst also minimising transaction and pricing administration costs.

A combination of various parameters is required in order to achieve economic functionality and to ensure that appropriate pricing signals are provided to customers. However, the number and design of these parameters has been selected with regard to minimising the associated transaction costs.

## 11.5 Response to price signals

### Regulatory Requirement

**Clause 6.18.5(b)(2)(ii) of the *Rules* states that:**

A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:

- (2) must be determined having regard to:
  - (ii) whether customers of the relevant *tariff class* are able or likely to respond to price signals.

### 11.5.1 Standard control services

Consistent with the *Rules*, the tariffs proposed by ENERGEX are comprised of a number of charging parameters that provide signals to customers about the efficient use of the network and the impact of future network capacity and costs. The charging parameters and their respective cost signals are discussed in detail in Section 6.2 for *standard control services*.



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The different types of charging parameters used in network pricing are:

- **capacity charges** – signals to network customers that they can reduce their network charges by reducing the variability of their consumption, as this lower peak demand reduces the pressure that they place on the network;
- **demand charges** – signals to network customers that they can reduce their charges by reducing their maximum demand, as it is levied on the basis that network customers who place greater pressure on the system should incur higher charges; and
- **volume charges** – by using a Time of Use structure (where metering allows) a volume charge can provide a small price signal that encourages more efficient use of the network by moving load from peak to off-peak times.

ENERGEX has designed these parameters with pricing signals that customers are able to respond to. ENERGEX is aware of customers putting systems in place to manage their network charges in response to these signals.

For example, commercial and industrial customers (who have these elements as part of the network tariff) may stagger the starting time of equipment such as motors, air-conditioning and even large lighting installations. This has the effect of reducing their maximum demand on the network, as all the start-up loads are not occurring at the same time. The short-term benefit to the customer is a lower monthly demand charge and if they effectively manage their longer term ATMD they will benefit from reduced capacity charges.

Energy usage costs are affected by the overall electrical efficiency of installations. If customers improve the efficiency of their usage this will reduce their energy related charges. For example, this can be achieved by installing more efficient lighting, improved thermal insulation of cold rooms and air-conditioned premises.

### 11.5.2 Alternative control services

Prices for *alternative control services* are calculated based on the current costs to provide a particular service. The pricing signals present are restricted to differentiating between services performed during business hours and those performed after hours. Customers are able to minimise the costs incurred for these services by choosing to have them delivered during business hours, where possible.

## 11.6 Tariff adjustment to address revenue shortfalls

### Regulatory Requirement

#### Clause 6.18.5(c) of the *Rules* states that:

If, however, as a result of the operation of paragraph 6.18.5(b), the *Distribution Network Service Provider* may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.



ENERGEX uses a combination of charging parameters in setting its network tariffs. These are developed taking into account the LRMC, transaction costs and customer response to pricing signals, as required under 6.18.5(b).

However, the expected revenue would not be recovered if charging parameters considered only those items identified in 6.18.5(b). Accordingly, other parameters are used in order to allow for the shortfall in expected revenue. These parameters are selected in a manner which complements the chosen pricing signals and minimises distortion.

## 11.7 Compliance with side constraints

### Regulatory Requirement

Clause 6.18.6 of the *Rules* states that:

- (a) This clause applies only to *tariff classes* related to the provision of *standard control services*.
- (b) The expected weighted average revenue to be raised from a *tariff class* for a particular *regulatory year* of a *regulatory control period* must not exceed the corresponding expected weighted average revenue for the preceding *regulatory year* by more than the permissible percentage.
- (c) The permissible percentage is the greater of the following:
  - (1) the CPI-X limitation on any increase in the *Distribution Network Service Provider's* expected weighted average revenue between the two *regulatory years* plus 2%;  
Note: The calculation is of the form  $(1 + \text{CPI})(1 - X)(1 + 2\%)$
  - (2) CPI plus 2%.  
Note: The calculation is of the form  $(1 + \text{CPI})(1 + 2\%)$
- (d) In deciding whether the permissible percentage has been exceeded in a particular *regulatory year*, the following are to be disregarded:
  - (1) the recovery of revenue to accommodate a variation to the distribution determination under rule 6.6 or 6.13;
  - (2) the recovery of revenue to accommodate pass through of charges for transmission use of system services to customers.
- (e) This clause does not, however, limit the extent a tariff for customers with remotely-read interval metering or other similar metering technology may vary according to the time or other circumstances of the customer's usage.

Although not a specific requirement under the *Rules* for the initial regulatory year, ENERGEX will demonstrate compliance with the side constraint formula for 2010-11.



In future annual pricing proposals, ENERGEX is required to demonstrate that the proposed DUOS tariffs for the next year ( $t$ ) will meet the following side constraints formula for each tariff class:

$$\frac{\sum_{j=1}^m d_t^j \times q_t^j}{\sum_{j=1}^m d_{t-1}^j \times q_t^j} \leq (1 + \Delta CPI_t) \times (1 - X_t) \times (1 + 2\%) \pm S_t \pm C_t \pm \text{transitional}_t \pm \text{passthrough}_t \pm \text{unders and overs}_t$$

where each tariff class ' $j$ ' has up to ' $m$ ' components, and where:

$d_t^j$  is the proposed price for component  $j$  of the tariff class for year  $t$

$d_{t-1}^j$  is the price charged by ENERGEX for component  $j$  of the tariff class in year  $t-1$

$q_t^j$  is the forecast quantity of component  $j$  of the tariff class in year  $t$

$\Delta CPI_t$  is the annual percentage change in the ABS Consumer Price Index All Groups, Weighted Average of Eight Capital Cities from March in regulatory year  $t-2$  to March in regulatory year  $t-1$

$X_t$  is the  $X$  factor for each year of the *regulatory control period*. If  $X > 0$ , then  $X$  will be set equal to zero for the purposes of the side constraint formula

$S_t$  is the Service Target Performance Incentive Scheme (STPIS) factor to be applied in regulatory year  $t$

$C_t$  is the annual adjustment factor for the difference between actual and forecast capital contributions in year  $t-2$

$\text{transitional}_t$  is a transitional factor for matters such as under/over in tax paid during the current *regulatory control period* and under/over adjustments related to shared assets used for purposes other than *standard control services*

$\text{passthrough}_t$  is an annual adjustment factor that reflects the pass through amounts approved by the AER with respect to regulatory year  $t$

$\text{unders \& overs}_t$  is an annual adjustment factor related to the balance of the DUOS under and over account with respect to regulatory year  $t$ .

In compliance with Clause 6.18.6(d) of the *Rules*, all TUOS charges and pass-throughs will be excluded from the calculation of the side constraint.

Table 11.4 demonstrates ENERGEX's compliance with the side constraint formula for each tariff class in 2010-11.

In addition, ENERGEX has applied the side constraint to individual customers in 2010-11. ENERGEX has side constrained a total of 11 customers, which results in a net impact on revenue of \$ 0.03m (as shown in Figure 4.1). Appendices 1.1 and 1.2 (confidential) provide details of the individual customers which are subject to the side constraint in 2010-11.



**Table 11.4 Compliance with side constraint formula**

| Tariff Class    | Weighted Average Revenue 2009-10 (\$m)<br>(regulatory year t-1) | Expected Revenue 2010-11 (\$m)<br>(regulatory year t, incl. side constraint adjustment) | % change (calculated) | % change (permissible) |
|-----------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------|------------------------|
| ICC             | 22.9                                                            | 25.2                                                                                    | 9.97%                 | 19.38%                 |
| CAC - 33kV      | 2.7                                                             | 2.9                                                                                     | 4.94%                 | 19.38%                 |
| CAC - 11kV Bus  | 11.8                                                            | 12.6                                                                                    | 6.79%                 | 19.38%                 |
| CAC - 11kV Line | 56.9                                                            | 62.3                                                                                    | 9.39%                 | 19.38%                 |
| EG              | 1.3                                                             | 1.3                                                                                     | 4.88%                 | 19.38%                 |
| SAC Demand      | 141.6                                                           | 164.5                                                                                   | 16.13%                | 19.38%                 |
| SAC Non-demand  | 716.5                                                           | 815.7                                                                                   | 13.84%                | 19.38%                 |

In order to calculate the % change for each tariff class, the weighted average revenue for 2009-10 in Table 11.4 is based on 2009-10 rates and 2010-11 quantities, as specified in the side constraint formula.

## 11.8 Summary of compliance with the *Rules* and distribution determination requirements

### Regulatory Requirement

**Clause 6.18.2(b)(7) of the *Rules* states that:**

*A pricing proposal must demonstrate compliance with the Rules and any applicable distribution determination.*

Table 11.5 demonstrates where the initial pricing proposal meets the requirements of each clause of Section 6.18 of the *Rules*.

**Table 11.5 Demonstrated compliance with the Rules**

| Clause       | Pricing Proposal Requirement                                                                                                                                                   | Reference              |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 6.1.4(a)     | Demonstrates that ENERGEX does not charge a Distribution Network User DUOS charges for the export of electricity generated by the user into the distribution network.          | Section 6.1            |
| 6.1.4(b)     | Demonstrates that where a Distribution Network User generates and exports electricity into the distribution network, ENERGEX charges for the provision of connection services. | Section 6.1            |
| 6.18.2(b)(1) | Sets out each tariff class (including the classes of <i>alternative control services</i> ).                                                                                    | Section 6<br>Section 7 |



| Clause                          | Pricing Proposal Requirement                                                                                                                                                                                                            | Reference                                                      |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 6.18.2(b)(2)                    | Sets out the proposed tariffs for each tariff class.                                                                                                                                                                                    | Table 6.4<br>Appendix 1<br>Appendix 2                          |
| 6.18.2(b)(3)                    | Set out for each proposed tariff, the charge parameters and the elements of service to which each charging parameter relates.                                                                                                           | Section 6.2<br>Section 7.1.3<br>Section 7.2.3<br>Section 7.3.3 |
| 6.18.2(b)(4)                    | Sets out the weighted average revenue for each tariff class related to <i>standard control services</i> .                                                                                                                               | Section 11.7                                                   |
| 6.18.2(b)(5)                    | Sets out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur.                                                                       | Section 8.1                                                    |
| 6.18.2(b)(6)                    | Sets out how TUOS will be passed through to customers.                                                                                                                                                                                  | Section 6.2.2<br>Section 9                                     |
| 6.18.2(b)(7)                    | Demonstrates compliance with the <i>Rules</i> and any applicable distribution determination.                                                                                                                                            | Section 11                                                     |
| 6.18.2(b)(8)                    | Describes the nature and extent of change from the previous regulatory year and demonstrates that the changes comply with the <i>Rules</i> and any applicable distribution determination.                                               | Section 8.2<br>Section 12                                      |
| 6.18.3(a)                       | Define the tariff classes into which customers for <i>direct control services</i> are divided.                                                                                                                                          | Section 5<br>Section 6.1<br>Section 7                          |
| 6.18.3(b)                       | Demonstrates that each customer for <i>direct control services</i> is a member of at least one tariff class.                                                                                                                            | Section 5                                                      |
| 6.18.3(c)                       | Sets out separate tariff classes for standard control and alternative control customers.                                                                                                                                                | Section 5<br>Section 6.1<br>Section 7                          |
| 6.18.3(d)(1)                    | Demonstrates that tariff classes are formed based on groupings of customers on an economically efficient basis.                                                                                                                         | Section 5                                                      |
| 6.18.3(d)(2)                    | Demonstrates that customers and tariffs are grouped into tariff classes with regard to the need to avoid unnecessary transactions costs.                                                                                                | Section 11.4                                                   |
| 6.18.4(a)(1)(i), (ii) and (iii) | Demonstrates that customers are assigned (or re-assigned) to tariff classes on the basis of the nature and extent of their usage, the nature of their connection to the network, and the metering installed at the customer's premises. | Section 5                                                      |
| 6.18.4(a)(2)                    | Demonstrates that customers with a similar usage and connection profile are treated on an equal basis.                                                                                                                                  | Section 5                                                      |
| 6.18.4(a)(3)                    | Demonstrates that customers with micro-generation facilities are treated no less favourably to customers without such facilities.                                                                                                       | Section 5.1                                                    |
| 6.18.4(a)(4), (b)               | Demonstrates that customer assignment (or reassignment) to a tariff class does not occur in the absence of an effective system of review.                                                                                               | Section 5.1.1                                                  |



| Clause               | Pricing Proposal Requirement                                                                                                                                                                                              | Reference                |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 6.18.5(a)(1) and (2) | Demonstrates that revenue from a tariff class lies between the stand-alone and incremental cost.                                                                                                                          | Section 11.2             |
| 6.18.5(b)(1)         | Demonstrates that tariffs and tariff components have regard for long run marginal cost.                                                                                                                                   | Section 11.3             |
| 6.18.5(b)(2)(i)      | Demonstrates that tariffs and tariff components have regard for the transaction costs to customers.                                                                                                                       | Section 11.4             |
| 6.18.5(b)(2)(ii)     | Demonstrates that tariffs and tariff components are set with regard to whether customers are able or likely to respond to signals.                                                                                        | Section 11.5             |
| 6.18.5(c)            | Demonstrates that if tariffs do not recover the required revenue as a result of the operation of clause 6.18.5(b), the tariffs have been adjusted with minimum distortion.                                                | Section 11.6             |
| 6.18.6(b)            | Demonstrates that the weighted average revenue for a standard control tariff class does not exceed that for the previous year by more than the “permissible percentage” defined in clause 6.18.6(c) of the <i>Rules</i> . | Section 11.7             |
| 6.18.6(c)(1) and (2) | Demonstrates the “permissible percentage” has been calculated in accordance with the definition set out in this clause of the <i>Rules</i> .                                                                              | Section 11.7             |
| 6.18.6(d)(1) and (2) | Demonstrates that TUOS and pass-throughs were removed from the calculation of the side constraint.                                                                                                                        | Section 11.7             |
| 6.18.6(e)            | Demonstrates that the side constraints have not impacted on the extent to which the tariffs for a customer with remotely read interval metering will vary according to usage.                                             | Section 11.7             |
| 6.18.7(a)            | Demonstrates that tariffs passed on to customers include the charges to be incurred by ENERGEX for TUOS services.                                                                                                         | Section 6.2<br>Section 9 |
| 6.18.7(b)            | Demonstrates that the TUOS charges passed on to customers do not exceed the estimated TUOS charges adjusted for over or under recovery in the previous regulatory year.                                                   | Section 9.3              |
| 6.18.7(c)(1) and (2) | Demonstrates that any TUOS over or under recovery is the difference between what was paid to Powerlink and what was recovered from customers via TUOS charges.                                                            | Section 9.3              |



| Clause       | Pricing Proposal Requirement                                                                                                                                                                             | Reference                                                                                                                 |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 6.18.9(a)(1) | The tariff classes and the applicable tariffs (or prices) are published on ENERGEX's website.                                                                                                            | The 2010-11 tariff schedule and statement of expected price trends will be published on the ENERGEX website in June 2010. |
| 6.18.9(a)(2) | The charging parameters within each tariff class (i.e. the fixed, demand and energy prices) and the elements of the service to which each charging parameter relates are published on ENERGEX's website. |                                                                                                                           |
| 6.18.9(a)(3) | The manner in which tariffs are expected to change over the remainder of the <i>regulatory control period</i> , and the reasons for the changes, are published on ENERGEX's website.                     |                                                                                                                           |
| 6.18.9(b)    | The information set out in clause 6.18.9(a) is published on ENERGEX's website 20 days prior to the start of the relevant regulatory year or as soon as practicable thereafter.                           |                                                                                                                           |

Table 11.6 demonstrates where the initial pricing proposal contains the details of any specific requirements under the Final Determination.

**Table 11.6 Demonstrated compliance with the requirements of the Final Determination**

| Section        | Pricing Proposal Requirement                                                                                                                                                               | Reference     |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Section 4.5.2  | Demonstrates that the proposed DUOS prices for the next year (t) will meet the side constraints formula set out in Section 4.5.2 of the Final Determination.                               | Section 11.7  |
| Section 4.6    | Include a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with Appendix E of the Final Determination.                                   | Section 9.3   |
| Section 17.3.3 | Include the revenues collected from the provision of each street light service in the preceding regulatory year.                                                                           | Appendix 2.4  |
| Section 18.3.5 | Include the volume of fee based and quoted services provided and the revenues recovered from the provision of fee based and quoted services in the preceding regulatory year.              | Appendix 2.4  |
| Section 2.5.2  | ENERGEX to follow the procedure for assigning customers to tariff classes or reassigning customers from one tariff class to another as specified in Appendix B of the Final Determination. | Section 5.1.1 |
| Appendix D     | Maintain a distribution use of system (DUOS) unders and overs account. ENERGEX must provide information on this account to the AER as part of its annual pricing proposals.                | Section 11.1  |
| Appendix E     | Maintain a transmission use of system (TUOS) unders and overs account. ENERGEX must provide information on this account to the AER as part of its annual pricing proposals.                | Section 9.3   |



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# 12 Customer impacts

## 12.1 Initial regulatory year (2010-11)

### 12.1.1 Standard control services

ENERGEX's pricing strategy is cognisant of the changing expectations of customers and the current upward pressure being exerted on energy prices. As an organisation ENERGEX is committed to achieving a balanced commercial outcome between meeting the requirements of customers and managing sustainability and risk.

The ongoing average price increases across the *regulatory control period* are necessary for ENERGEX to deliver the capital and operating expenditure programs which support development and growth in SEQ, as well as improving reliability and security of supply.

Table 12.1 provides an estimate of the charges in 2010-11 for the average consumption level in each tariff class. This table is used to give an estimate of the percentage change for the average customer. The impact on each customer will be dependent on the individual customer's demand and consumption patterns.

**Table 12.1 Outline of estimated % price change by tariff class for 2009-10 to 2010-11**

| Tariff Class        | Estimated DUOS charge 2009-10 (c/kW.h) | Estimated DUOS charge 2010-11 (c/kW.h) | Percentage change (%) |
|---------------------|----------------------------------------|----------------------------------------|-----------------------|
| ICC                 | 1.30                                   | 1.43                                   | 9.97%                 |
| CAC – 33kV Line/Bus | 1.85                                   | 1.94                                   | 4.94%                 |
| CAC – 11kV Bus      | 1.50                                   | 1.60                                   | 6.79%                 |
| CAC – 11kV Line     | 1.79                                   | 1.96                                   | 9.39%                 |
| EG                  | 2.72                                   | 2.85                                   | 4.88%                 |
| SAC Demand          | 3.10                                   | 3.61                                   | 16.13%                |
| SAC Non-demand      | 6.00                                   | 6.83                                   | 13.84%                |

### 12.1.2 Alternative control services

The price changes between 2009-10 and 2010-11 for *alternative control services* (street lighting and fee based services) are provided in Table 12.2 and Table 12.3. These first year price changes were approved by the AER in the Final Determination.



**Table 12.2 Street light services % increase for 2009-10 to 2010-11**

| <b>Street lighting service</b> | <b>% change<br/>(2009-10 to 2010-11)</b> |
|--------------------------------|------------------------------------------|
| Major non-contributed          | -11.16%                                  |
| Major contributed              | -75.64%                                  |
| Minor non-contributed          | 39.55%                                   |
| Minor contributed              | -41.47%                                  |

**Table 12.3 Fee based services % increase for 2009-10 to 2010-11**

| <b>Fee based Service</b>                                                     | <b>% change<br/>(2009-10 to 2010-11)</b> |
|------------------------------------------------------------------------------|------------------------------------------|
| Alterations and additions to current metering equipment                      | -33.83%                                  |
| Attending loss of supply – LV customer installation at fault (BH)            | -38.42%                                  |
| Overhead service replacement – single phase                                  | 26.16%                                   |
| Overhead service replacement – multiple phase                                | 19.18%                                   |
| De-energisation                                                              | -17.19%                                  |
| Meter test                                                                   | -23.43%                                  |
| Meter inspection                                                             | 0.00%                                    |
| Reconfigure meter                                                            | 22.21%                                   |
| Off-cycle meter read                                                         | -74.16%                                  |
| Site visit                                                                   | 6.30%                                    |
| Locating ENERGEX underground cables                                          | -9.70%                                   |
| Temporary connection                                                         | 22.53%                                   |
| Re-energisation – business hours                                             | -45.85%                                  |
| Re-energisation – after hours                                                | -6.25%                                   |
| Re-energisation (visual) – business hours                                    | -7.94%                                   |
| Re-energisation (visual) – after hours                                       | 22.48%                                   |
| Re-energisation non-payment (visual) – business hours                        | -7.94%                                   |
| Re-energisation non-payment (visual) – after hours                           | 22.48%                                   |
| Supply abolishment                                                           | 185.92%                                  |
| Unmetered supply                                                             | -49.45%                                  |
| Street light glare screening                                                 | 3.34%                                    |
| Replacement of standard luminaries with aero screen units (per street light) | 1.01%                                    |



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## 12.2 Expected price trends (2010-11 to 2014-15)

### Regulatory Requirement

**Clause 6.18.9(a)(3) states that ENERGEX must maintain on its website:**  
a statement of expected price trends (to be updated for each *regulatory year*) giving an indication of how the *Distribution Network Service Provider* expects prices to change over the *regulatory control period* and the reasons for the expected changes.

A statement of the expected price trends for 2010-11 to 2014-15 will be published annually on the ENERGEX website.

In summary,

- For *standard control services*, any change in prices will be subject to the approved X factors from the Final Determination, the change in CPI and any yearly adjustments which are applicable.
- In each regulatory year (as part of the requirements of a pricing proposal under the *Rules*), ENERGEX will demonstrate that the proposed tariffs for DUOS will meet the side constraints formula for each tariff class.
- For *alternative control services* including street lighting and fee based services, the prices will be subject to an annual escalation process.



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## 13 Annual approval process

ENERGEX will submit annually to the AER a schedule of proposed tariffs that are to apply in the following financial year. ENERGEX will demonstrate that the proposed tariffs comply with the *Rules* and the requirements of the AER Final Determination.

For the initial pricing proposal, the dates for the annual approval process are:

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30 April 2010 | <p>The AER must as soon as practicable, but not later than 2 months before the commencement of the relevant regulatory control period, publish:</p> <ol style="list-style-type: none"><li>(1) notice of the making of the distribution determination; and</li><li>(2) the distribution determination itself; and</li><li>(3) the AER's reasons for making the distribution determination in its final form including the constituent decisions i.e. the decisions made in accordance with rule 6.12 on which the distribution determination is predicated.</li></ol> <p>On 30 March 2010 the AER advised that it will now publish the Final Determination on 6 May 2010. An extension is required to incorporate into modelling the March quarter consumer price index to be released by the Australian Bureau of Statistics on 28 April 2010.</p> |
| 27 May 2010   | <p>ENERGEX submits to the AER, as soon as practicable, and in any case within 15 business days, after publication of the distribution determination, a pricing proposal (the "initial pricing proposal") for the first regulatory year of the <i>regulatory control period</i>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| June 2010     | <p>Approval from the AER if satisfied that the proposed prices comply with the <i>Rules</i> and the requirements under the determination.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| June 2010     | <p>ENERGEX publishes price schedule for the coming financial year on ENERGEX's website 20 business days prior to the start of the relevant regulatory year or as soon as practicable thereafter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



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For subsequent pricing proposals (in 2011, 2012, 2013 and 2014), the dates for the annual approval process are:

|          |                                                                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30 April | ENERGEX submits to the AER, at least 2 months before the commencement of the second and each subsequent regulatory year of the <i>regulatory control period</i> , a further pricing proposal (an "annual pricing proposal") for the relevant regulatory year. |
| May      | Approval from the AER if satisfied that the proposed prices comply with the <i>Rules</i> and the requirements under the determination.                                                                                                                        |
| June     | ENERGEX publishes price schedule for the coming financial year on ENERGEX's website 20 business days prior to the start of the relevant regulatory year or as soon as practicable thereafter.                                                                 |



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# 14 Glossary

## 14.1 Abbreviations

| Term   | Definition                            |
|--------|---------------------------------------|
| AEMO   | Australian Energy Market Operator     |
| AER    | Australian Energy Regulator           |
| AIC    | Average Incremental Cost              |
| ATMD   | Anytime Maximum Demand                |
| B2B    | Business-to-business                  |
| c/kW.h | cents per Kilowatt hour               |
| CAB    | Contributed Asset Base                |
| CAC    | Connection Asset Customers            |
| CAM    | Cost Allocation Method                |
| CPI    | Consumer Price Index                  |
| DCOS   | Distribution Cost of Supply           |
| DME    | Department of Mines and Energy        |
| DNSP   | Distribution Network Service Provider |
| DUOS   | Distribution Use of System            |
| EG     | Embedded Generators                   |
| EUAA   | Energy Users Association of Australia |
| GW.h   | Gigawatt hour                         |
| HV     | High Voltage                          |
| ICC    | Individually Calculated Customers     |
| kV     | Kilovolt                              |
| kV.A   | Kilovolt Ampere                       |
| kW.h   | Kilowatt hour                         |
| LRMC   | Long Run Marginal Cost                |
| LV     | Low Voltage                           |
| MAR    | Maximum Allowable Revenue             |
| MW     | Mega Watt                             |
| NEL    | National Electricity Law              |
| NEM    | National Electricity Market           |



| Term  | Definition                                 |
|-------|--------------------------------------------|
| NMI   | National Meter Identifier                  |
| NUOS  | Network Use of System                      |
| O&M   | Operating and Maintenance Allowance (Opex) |
| PTRM  | Post Tax Revenue Model                     |
| PV    | Photovoltaic                               |
| QCA   | Queensland Competition Authority           |
| RAB   | Regulatory Asset Base                      |
| Rules | National Electricity Rules                 |
| SAC   | Standard Asset Customers                   |
| SEQ   | South East Queensland                      |
| TNCP  | Transmission Network Connection Point      |
| TNSP  | Transmission Network Service Provider      |
| TOU   | Time of Use                                |
| TUOS  | Transmission Use of System                 |
| WACC  | Weighted Average Cost of Capital           |

## 14.2 Definitions

| Term                          | Definition                                                                                                                                                                                                                                                                                                             |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alternative Control           | This service class includes the provision, construction and maintenance of street lighting assets, and fee based and quoted services.                                                                                                                                                                                  |
| Anytime Maximum Demand (ATMD) | Anytime Maximum Demand is the maximum demand recorded at anytime. Can be used to indicate the maximum demand that occurs anytime within a specified time period such as a 12 calendar month period or a specific month.                                                                                                |
| Capacity charge               | This part of the tariff seeks to reflect the costs associated with providing network capacity required by a customer on a long-term basis. It is levied on the basis of either contracted demand or the maximum demand in the previous calendar year. The charge is applied as a fixed dollar amount per kW per month. |
| Capacity-Network              | The maximum demand (kW) that the distribution network can provide for at any one time.                                                                                                                                                                                                                                 |
| Charging Parameter            | The constituent elements of a tariff (as per the Rules definition).                                                                                                                                                                                                                                                    |
| Common Service                | A service that ensures the integrity of a distribution system and benefits all Distribution Customers and cannot reasonably be allocated on a locational basis                                                                                                                                                         |



| Term                                              | Definition                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Connection Asset Customer (CAC)                   | Typically those customers with electricity consumption greater than 4GW.h (but less than 40GW.h) per year at a single connection point; or where a customer has a dedicated supply system with significant connection assets.                                                                 |
| Connection Asset (Contributed or Non-contributed) | Related to building connection assets at a customer's premises as well as the connection of these assets to the distribution network.<br><br>Connection assets can be contributed (customer funded, then gifted to ENERGEX) or non-contributed (ENERGEX funded).                              |
| Connection Point                                  | The point of electrical coupling between the electricity distribution network and a customer's electrical installation. The meter is installed as close a possible to this location.                                                                                                          |
| Controlled Load 1                                 | Specified permanently connected appliances are controlled by network equipment so that supply will be permanently available for a minimum period of 8 hours at the absolute discretion of ENERGEX but usually between the hours of 10:00pm and 7:00am.                                        |
| Controlled Load 2                                 | Specified connected appliances are controlled by network equipment so that supply will be available for a minimum period of 18hours per day during time periods set at the absolute discretion of ENERGEX.                                                                                    |
| Demand                                            | The amount of electricity energy being consumed at a given time measured in either watts (W) or volt amperes (VA). The difference between the two is the power factor.                                                                                                                        |
| Demand Metered SAC                                | The customer's connection point has a meter installed that is capable of measuring energy consumption (kW.h) and demand (kW). This meter records total energy consumption (kW.h) and demand over 30 minute periods. A customer's demand is the average demand (kW) over the 30 minute period. |
| Demand Metered Tariff                             | The tariff has been built to include a demand component so that the customer's actual demand is reflected in the price that they pay for their electricity. The highest demand reading for that month is used to calculate the customer's electricity bill.                                   |
| Demand Charge                                     | This part of the tariff accounts for the actual demand that a customer places on the electricity network. The actual demand levied for billing purposes is the metered monthly maximum demand. The charge is applied as a fixed dollar per kW per month.                                      |
| Distribution Cost of Supply (DCOS) Model          | The ENERGEX model used to allocate costs approved by the AER to the various tariff classes.                                                                                                                                                                                                   |
| Distribution Loss Factor                          | Distribution Loss Factors (DLFs) represent the average electrical energy losses incurred when electricity is transmitted over a distribution network.                                                                                                                                         |
| Distribution Use of System (DUOS)                 | This refers to the network charges for the use of the distribution network.                                                                                                                                                                                                                   |



| Term                                   | Definition                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Embedded Generator                     | In line with the Energy Networks Association (ENA) classification, Embedded Generators (EGs) are generally those generators with an installed capacity as follows:<br>Medium: 1-5 MW (LV or HV) or <1MW (HV)<br>Large: >5MW                                                                                                                                                  |
| Energy                                 | The amount of electricity consumed by a consumer (or all customers) over a period of time. Energy is measured in terms of watt hours (W.h), kilowatt hours (kW.h), megawatt hours (MW.h) or gigawatt hours (GW.h).                                                                                                                                                           |
| Energy Charge                          | This part of the tariff seeks to reflect costs not directly allocated to network drivers and costs that are proportional to the size of the customer. The energy consumption (kW.h) for the period, as recorded by the customer's meter, is utilised to calculate this part of the tariff charge. This charge is applied as a fixed amount (cents) per Kilowatt hour (kW.h). |
| Energy Charge (Off-peak)               | This charge is applicable to those customers who are on a Time of Use tariff. The energy consumption (kW.h) during Off-peak periods (refer to Off-peak Period for Off-peak Period times), as recorded by the customer's meter, is utilised to calculate this part of the tariff. This charge is applied as a fixed amount (cents) per Kilowatt hour (kW.h).                  |
| Energy Charge (Peak)                   | This charge is applicable to those customers who are on a Time of Use tariff. The energy consumption (kW.h) during Peak periods (refer to Peak Period for Peak Period times), as recorded by the customer's meter, is utilised to calculate this part of the tariff. This charge is applied as a fixed amount (cents) per Kilowatt hour (kW.h).                              |
| Fixed Charge                           | The fixed charge seeks to reflect the costs associated with customer's dedicated connection assets. The charge is applied as a fixed dollar amount per day.                                                                                                                                                                                                                  |
| High Voltage (HV)                      | Refers to the 11kV or above network.                                                                                                                                                                                                                                                                                                                                         |
| Individually Calculated Customer (ICC) | Typically those customers with electricity consumption greater than 40GW.h per year at a single connection point; or where a customer's circumstances mean that the average shared network charge becomes meaningless or distorted.                                                                                                                                          |
| kV.A                                   | Kilovolt Amperes, which is a measure of the apparent power being consumed and is used to measure demand. One kV.A equals 1,000 Volt Amperes.                                                                                                                                                                                                                                 |
| Low Voltage (LV)                       | Refers to the sub-11 kV network                                                                                                                                                                                                                                                                                                                                              |
| Maximum Allowable Revenue              | The maximum revenue which can be recovered through tariffs for the regulatory year.                                                                                                                                                                                                                                                                                          |
| Maximum Demand                         | The maximum demand recorded at a customer's individual meter or the maximum demand placed on the electrical distribution network system at any time or at a specific time or within a specific time period such as a month. Maximum demand is an indication of the capacity required for a customer's connection or the electrical distribution network.                     |



| Term                                                                  | Definition                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Micro Generator                                                       | AS4777 compliant generators with an installation size of less than 10kW (single phase) or 30kW (three phase) connected to the LV network.                                                                                                           |
| Minimum Chargeable Demand                                             | The minimum chargeable demand will be applied to customers, for billing purposes, in each month where the metered monthly maximum demand fails to exceed the minimum chargeable demand for their tariff code.                                       |
| National Electricity Market (NEM)                                     | The interconnected electricity grid covering Queensland, New South Wales, Victoria, Tasmania, South Australia and the Australian Capital Territory.                                                                                                 |
| National Metering Identifier (NMI)                                    | A unique number assigned to each metering installation.                                                                                                                                                                                             |
| Network Use of System (NUOS)                                          | The tariff for use of the distribution and transmission networks. It is the sum of both Distribution Use of System and Transmission Use of System.                                                                                                  |
| Non-demand Metered SAC                                                | The customer's connection point has a meter installed that is capable of measuring the total energy consumption (kW.h).                                                                                                                             |
| Non-demand Metered Tariff                                             | The tariff is based around a fixed daily component and the actual energy (kW.h) used by the customer.                                                                                                                                               |
| Off-peak Period                                                       | All hours which are outside of Peak Period hours.                                                                                                                                                                                                   |
| Peak Period                                                           | Meter type 1-4 (ICC, CAC & SAC demand): The hours between 7am and 11pm, Monday to Friday. Meter type 6 (SAC Non-demand): The hours between 7am and 9pm, Monday to Friday.                                                                           |
| Power Factor                                                          | Power factor, is the ratio of kW to kV.A, and is a useful measure of the efficiency in the use of the network infrastructure. The closer to one the power factor, the more efficient the network assets are utilised.<br><br>Power Factor = kW/kV.A |
| Price Path                                                            | Outlines the escalation factors to be applied to the initial price over the <i>regulatory control period</i>                                                                                                                                        |
| Queensland Government Solar Bonus Scheme for Standard Asset Customers | A program that pays domestic and other small energy customers for the surplus electricity generated from roof-top solar photovoltaic (PV) systems that is exported to the Queensland grid.                                                          |
| Regulatory Depreciation                                               | Associated with recouping the capital that has been invested in the business assets over the useful lives of those assets.                                                                                                                          |
| Return on Capital                                                     | The return necessary to achieve a fair and reasonable rate of return on the assets necessarily invested in the business.                                                                                                                            |
| Site Specific Charge                                                  | This charge is calculated specifically for a site and is specific to the individual connection point.                                                                                                                                               |
| Small generator                                                       | AS4777 compliant generators with an installation size of less than 1MW connected to the LV network.                                                                                                                                                 |



| Term                              | Definition                                                                                                                                                                                   |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solar Photovoltaic (Solar PV)     | A Solar Photovoltaic system uses sunlight to generate electricity for domestic use. The system provides power for the premises with any excess production feeding into the electricity grid. |
| Standard Asset Customer (SAC)     | Generally those customers with an annual electricity consumption below 4GW.h per year, whose supply arrangements are consistent across the customer group.                                   |
| Standard Control                  | This service class includes network, connection and metering services.                                                                                                                       |
| Street lights (Major)             | Larger, higher wattage street lights typically used on major roads and are categorised according to lamp size, as per attachment in Appendix 5.                                              |
| Street lights (Minor)             | Smaller, lower wattage street lights typically used on minor roads and are categorised according to lamp size, as per attachment in Appendix 5.                                              |
| Tariff class                      | A class of customers for one or more <i>direct control services</i> who are subject to a particular tariff or particular tariffs (as per the Rules definition).                              |
| Time of Use (ToU)                 | Refers to tariffs that vary according to the time of day at which the electricity is consumed.                                                                                               |
| Transmission Use of System (TUOS) | Refers to the charges incurred for use of the transmission network.                                                                                                                          |
| Unmetered Supply                  | A customer who takes supply where no meter is installed at the connection point.                                                                                                             |



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# 15 Confidential information

## Claim for confidentiality

ENERGEX claims confidentiality over the following sections and appendices on the grounds that the information is either Commercial-in-confidence or contains intellectual property. ENERGEX requests that the AER does not disclose the information contained in these sections to any person outside of the AER:

### Appendices

| No. | Title                                                                     |
|-----|---------------------------------------------------------------------------|
| 1.1 | ICC site specific tariffs                                                 |
| 1.2 | CAC and EG site specific tariffs                                          |
| 2.4 | Additional reporting requirements for <i>alternative control services</i> |
| 4.0 | Reclassification of customers to tariff classes                           |



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## 16 Appendices

| No. | Title                                                                                                                                                                                                                                                          |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | 2010-11 Tariffs – <i>Standard Control Services</i><br>1.1 ICC site specific tariffs (Confidential)<br>1.2 CAC and EG site specific tariffs (Confidential)<br>1.3 kV.A calculation methodology for ICC and CAC customers                                        |
| 2   | 2010-11 Tariffs – <i>Alternative Control Services</i><br>2.1 Street lighting services<br>2.2 Fee-based services<br>2.3 Quoted services – Representative sample<br>2.4 Additional reporting requirements for <i>alternative control services</i> (Confidential) |
| 3   | Tariff cost allocation process                                                                                                                                                                                                                                 |
| 4   | Reclassification of customers to tariff classes (Confidential)                                                                                                                                                                                                 |
| 5   | Street light classification                                                                                                                                                                                                                                    |
| 6   | In-period price review process                                                                                                                                                                                                                                 |
| 7   | Capital contributions policy                                                                                                                                                                                                                                   |



## **Appendix 1**

### **2010-11 Network Tariffs Standard Control Services**

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#### **Contents**

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**1.1 ICC site specific tariffs (confidential)**

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**1.2 CAC and EG site specific tariffs (confidential)**

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**1.3 kV.A calculation methodology for ICC and CAC customers**

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## **1.1** | ICC site specific tariffs (confidential)

## Appendix 1.1

**CONFIDENTIAL**

### **2010-11 Tariffs Standard Control Services ICC**





[illegible]

## **1.2** | CAC and EG site specific tariffs (confidential)

## Appendix 1.2

**CONFIDENTIAL**

### **2010-11 Tariffs Standard Control Services CAC and EG Site Specific Tariffs**



**Table 1 Site specific tariffs for CAC's (2010-11)**













































Table 2 Site specific tariffs for EG's (2010-11)



## **1.3** | **kV.A calculation methodology for ICC and CAC customers**

## Appendix 1.3

# 2010-11 Tariffs Standard Control Services kV.A calculation methodology for ICC and CAC customers

From 1 July 2010, charges for ICC's and CAC's will be adjusted up or down by a relative amount based on the difference between the customers' measured power factor and the power factor neutral point. The adjustment will be applied to the demand and capacity charge elements as outlined below.

### 1. kV.A Tariff Adjustment Calculation Methodology

a. 
$$\text{Actual Power Factor (APF)} = \frac{\text{Maximum monthly kW}}{\text{Maximum monthly kV.A}}$$

where monthly kW and monthly kV.A data is collected from the customer's meter

b. 
$$\text{Power Factor Adjustment (PFA)} = \text{Multiplier} \times (\text{TPF} - \text{APF}) \times (1 + \text{TPF} - \text{APF})^{\text{index}}$$

where TPF = Target Power Factor (Power factor neutral point of 0.9 for ICCs, 0.85 for CACs)

Multiplier and index are variables set by ENERGEX (as per table below)

The PFA will be limited to the maximum and minimum bounds defined by ENERGEX as required

c. 
$$\text{Demand Adjustment (kW)} = \text{PFA} \times \text{Maximum monthly demand}$$

$$\text{Capacity Adjustment (kW)} = \text{PFA} \times \text{Nominated monthly demand}$$

The adjustments (measured in kW) will be positive or negative depending on the difference between the customers actual power factor (APF) and target power factor (TPF). This will result in an increase or decrease to the monthly charge as shown in the formula below.

d. 
$$\text{Monthly charge for DUOS(\$)} = \underbrace{\text{Fixed} + \text{Demand} + \text{Capacity} + \text{Peak Energy} + \text{Off Peak Energy}}_{\text{Existing parameters}} + \underbrace{\text{Demand Adjustment} + \text{Capacity Adjustment}}_{\text{New kV.A adjustments for 2010/11}}$$

where Demand Adjustment (\$) = PFA x Maximum monthly demand (kW) x Demand Charge (\$/kW/month)

Capacity Adjustment (\$) = PFA x Nominated monthly demand (kW) x Capacity Charge (\$/kW/month)



## 2. kV.A Tariff Adjustment Parameters for 2010-11

| Customer Type | Target Power Factor | Index   | Multiplier | Maximum Adjustment | Minimum Adjustment |
|---------------|---------------------|---------|------------|--------------------|--------------------|
| ICC - 110B    | 0.9000              | -1.9445 | 0.5000     | 1.00%              | -10.00%            |
| ICC - 11B     | 0.9000              | -2.9393 | 0.8000     | 5.00%              | -10.00%            |
| ICC - 33B     | 0.9000              | -2.9393 | 0.8000     | 5.00%              | -10.00%            |
| CAC - 11B     | 0.8500              | -1.1934 | 0.7500     | 10.00%             | -10.00%            |
| CAC - 11L     | 0.8500              | 1.7131  | 0.5000     | 10.00%             | -5.00%             |
| CAC - 33L     | 0.8500              | 0.5838  | 0.8000     | 5.00%              | -10.00%            |



### 3. kV.A Tariff Adjustment Example

The following example is used to illustrate the application of the kV.A tariff adjustment for a customer of the CAC 11kV Line tariff class.

| Input Parameter     | Value                  | Reference                                        |
|---------------------|------------------------|--------------------------------------------------|
| Tariff Class        | CAC 11kV Line          |                                                  |
| Nominated kW Demand | 2000kW                 |                                                  |
| Maximum kW Demand   | 1800kW                 | Collected from customer's meter data             |
| Maximum kVA Demand  | 2000kW                 | Collected from customer's meter data             |
| Multiplier          | 0.5                    | As specified for tariff class CAC 11 Line        |
| Index               | 1.7131                 | As specified for tariff class CAC 11 Line        |
| Maximum adjustment  | 10%                    | As specified for tariff class CAC 11 Line        |
| Minimum adjustment  | -5%                    | As specified for tariff class CAC 11 Line        |
| Target Power Factor | 0.85                   | As specified for CAC tariff class                |
| Demand Charge       | 4.20031<br>\$/kW/month | As proposed for 2010-11 in Table 6.4 (DUOS only) |
| Capacity Charge     | 2.04251<br>\$/kW/month | As proposed for 2010-11 in Table 6.4 (DUOS only) |

a. Actual Power Factor (APF) =  $\frac{\text{Maximum monthly kW}}{\text{Maximum monthly kV.A}} = \frac{1800}{2000} = 0.900$

b. Power Factor Adjustment (PFA) =  $\text{Multiplier} \times (\text{TPF} - \text{APF}) \times (1 + \text{TPF} - \text{APF})^{\text{index}}$   
 $= 0.5 \times (0.85 - 0.900) \times (1 + 0.85 - 0.900)^{1.7131}$   
 $= -0.0229$   
 $= -2.29\%$

The calculated PFA is between the minimum and maximum adjustment bounds, so the limiting boundaries do not need to be applied.

c. Demand Adjustment (kW) =  $\text{PFA} \times \text{Maximum monthly demand}$   
 $= -0.0229 \times 1800$   
 $= -41\text{kW}$

Capacity Adjustment (kW) =  $\text{PFA} \times \text{Nominated monthly demand}$   
 $= -0.0229 \times 2000$   
 $= -46\text{ kW}$

d. Demand Adjustment (\$) =  $\text{PFA} \times \text{Maximum Monthly Demand (kW)} \times \text{Demand Charge (\$/kW/month)}$   
 $= -0.0229 \times 1800 \times 4.20031$   
 $= -\$173.14$

Capacity Adjustment (\$) =  $\text{PFA} \times \text{Nominated Monthly Demand (kW)} \times \text{Capacity Charge (\$/kW/month)}$   
 $= -0.0229 \times 2000 \times 2.04251$   
 $= -\$93.55$



## **Appendix 2**

### **2010-11 Network Tariffs Alternative Control Services**

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#### **Contents**

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**2.1 Street lighting services**

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**2.2 Fee-based services**

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**2.3 Quoted services**

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**2.4 Additional reporting requirements for alternative control services  
(Confidential)**

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## **2.1** | Street lighting services

## Appendix 2.1

### 2010-11 Tariffs Alternative Control Services Street lighting services

The approved prices for the provision, construction and maintenance of street light services are provided in Table 1.

**Table 1**      **ENERGEX prices for street lighting services**

| <b>Streetlight Service</b>       | <b>2010–11<br/>Price<br/>(\$ per light per day)</b> |
|----------------------------------|-----------------------------------------------------|
| Major non-contributed            | 0.85                                                |
| Major contributed                | 0.23                                                |
| Minor non-contributed            | 0.34                                                |
| Minor contributed                | 0.09                                                |
| All prices are exclusive of GST. |                                                     |



## 2.2 | Fee-based services

## Appendix 2.2

### 2010-11 Tariffs Alternative Control Services Fee-based services

The AER approved prices for individual fee-based services provided by ENERGEX are outlined in Table 1 below. These prices were determined using the AER's approved formula based price cap control mechanisms.

**Table 1** Prices for fee-based services for the *2010-11 regulatory year*

| Fee-based Service                                                            | 2010–11 Price<br>(\$ per service) |
|------------------------------------------------------------------------------|-----------------------------------|
| Alterations and additions to current metering equipment                      | 89.85                             |
| Attending loss of supply – LV customer installation at fault (BH)            | 100.36                            |
| Overhead service replacement – single phase                                  | 274.76                            |
| Overhead service replacement – multiple phase                                | 323.67                            |
| De-energisation                                                              | 45.00                             |
| Meter test                                                                   | 103.98                            |
| Meter inspection                                                             | 80.31                             |
| Reconfigure meter                                                            | 66.58                             |
| Off-cycle meter read                                                         | 7.50                              |
| Site visit                                                                   | 57.67                             |
| Locating ENERGEX underground cables                                          | 122.62                            |
| Temporary connection                                                         | 798.91                            |
| Re-energisation – business hours                                             | 39.21                             |
| Re-energisation – after hours                                                | 111.89                            |
| Re-energisation (visual) – business hours                                    | 66.66                             |
| Re-energisation (visual) – after hours                                       | 146.18                            |
| Re-energisation non-payment (visual) – business hours                        | 66.66                             |
| Re-energisation non-payment (visual) – after hours                           | 146.18                            |
| Supply abolishment                                                           | 311.19                            |
| Unmetered supply                                                             | 137.34                            |
| Street light glare screening                                                 | 132.73                            |
| Replacement of standard luminaires with aero screen units (per street light) | 306.85                            |

All prices are exclusive of GST.  
Business Hours (BH); After Hours (AH).



## 2.3 | Quoted services

## Appendix 2.3

### 2010-11 Tariffs Alternative Control Services Quoted services

Table 1 sets out the AER's approved prices for ENERGEX's quoted services. These prices are based on the illustrative service configuration and were determined using the AER's approved formula based price cap control mechanisms. The AER's approved prices do not represent a binding capped price for an individual quoted service.

**Table 1 Approved prices for ENERGEX's quoted services (illustrative configurations)**

| Quoted Service                                                         | 2010–11 Price<br>(\$ per service) |
|------------------------------------------------------------------------|-----------------------------------|
| Rearrangement of network assets                                        | 3,801.93                          |
| Customer requested works to allow customer or contractor to work close | 5,702.62                          |
| Non standard data and metering services (type 5-7 metering)            | 101.59                            |
| Emergency recoverable works and rectification of illegal connections   | 8,454.12                          |
| Large customer connections                                             | 321,662.10                        |
| Design specification/auditing and other subdivision activities         | 1,219.06                          |
| Unmetered services, including street lighting                          | 1,647.04                          |
| After hours provision of any fee based service (excl re-energisation)  | 1,471.48                          |
| Supply abolishment - complex                                           | 404.31                            |
| Additional crew                                                        | 107.14                            |
| Temporary connection - complex                                         | 41,812.20                         |
| Loss of asset                                                          | 8,514.70                          |
| Other recoverable work <sup>1</sup>                                    | n/a                               |

1. There is no common configuration of the 'other recoverable work' service. The service is applied only in those circumstances where the service requested is not covered by any of the other service categories or would not otherwise have been requested for the efficient management of the network.

All indicative prices are exclusive of GST



## **2.4** | **Additional reporting requirements for alternative control services (confidential)**

## Appendix 2.4

**CONFIDENTIAL**

## Additional Reporting Requirements for Alternative Control Services

| Device Type       | Percentage of Respondents |
|-------------------|---------------------------|
| Smartphone        | 100%                      |
| Tablet            | 100%                      |
| Laptop            | 100%                      |
| Desktop Computer  | 100%                      |
| Smart TV          | 100%                      |
| Smartwatch        | 100%                      |
| Smart Speaker     | 100%                      |
| Smart Car         | 100%                      |
| Smart Home System | 100%                      |
| Smart Lock        | 100%                      |

**Table 1      Revenue collected from street light services 2008-09**

| Streetlight Service | Revenue Collected<br>2008-09<br>(\$m) |
|---------------------|---------------------------------------|
|                     |                                       |
|                     |                                       |
|                     |                                       |
|                     |                                       |

**Table 2      Volume of fee based services and revenue collected for previous regulatory year**

[illegible]

Table 3 Volume of quoted services and revenue collected for previous regulatory year

[illegible]

## **Appendix 3**

### **Tariff Cost Allocation Process**

## Appendix 3

# Tariff cost allocation process

The revenue cap as determined by the AER is based on a building block approach, which includes each of the regulated cost components, namely:

- operating expenditure;
- regulatory depreciation;
- return on capital; and
- tax allowance.

The first stage of the tariff determination process is to allocate or assign the network costs to the tariff classes in the most efficient and cost reflective way. That is, costs are allocated on a causal basis. This section sets out the cost allocation process utilised by ENERGEX in the determination of tariffs.

## 1 Operating Expenditure

The AER has determined an overall efficient operating expenditure target for ENERGEX as part of the regulatory review process. To progress through the cost allocation process, this overall target is separated into a number of cost groups that are consistent in relation to drivers as specified in the Post Tax Revenue Model (PTRM). These operating expenditure cost groups are:

- **direct** – the directly attributable costs associated with the provision of network connection and distribution services;
- **oncosts** – expenditure items that do not meet the definition of direct operating expenditure but arise as a consequence of incurring direct costs; and
- **overheads and other operating costs** – costs including corporate support (e.g. CEO, Finance, Human Resources, Legal), customer services, and occupancy costs which are not directly attributable to the operation and maintenance of the network but which are associated with network service delivery.

Operating expenditure in the PTRM is allocated in accordance with the approved Cost Allocation Methodology (CAM).

### 1.1 Direct Operating Expenditure

Direct operating expenditure costs are sub-classified into two categories, Network and Common Services, based on their written down asset values.

Network costs are those costs directly associated with the operation and maintenance of asset classes that are attributable to a single customer or group of customers.

The Distribution Cost of Supply (DCOS) model is used to allocate these network costs to the users of the respective asset class. For example, the customers taking supply at the 11kV level will be



allocated a proportion of all of the upstream costs (33kV, 110kV, etc) while those taking supply at the 110kV level will be allocated their proportion of the 110kV system costs only (not downstream system costs as they do not use those parts of the system).

The voltage level direct network operating expenditure cost groups are:

- 110/132kV;
- 33kV;
- 11kV Bus;
- 11kV Line;
- low voltage;
- services; and
- meters.

Direct operating expenditure is segregated into the voltage level cost groups based, where possible, on the current forecast of direct operating and maintenance expenditure for the asset classes. Where this data is not available, the replacement costs of assets are used to allocate direct operating expenditure.

Common Services costs are costs associated with those system assets that benefit the system as a whole and are not directly related to any single customer or group of customers. Assets included in this category are reactive plant, load control, control centres and communications.

## 1.2 On-costs

On-costs are costs that are associated with direct costs and include labour, materials storage and logistics and fleet. These costs will be attributed directly to the above listed voltage levels in proportion to the allocated direct costs.

## 1.3 Overheads and other operating costs

Overheads and other operating costs are the summation of the non-system based costs which include corporate overheads and customer services. These costs are treated consistently as a group as it is impractical to manage a cost allocation stream for each of the specific components, for example, CEO, IT&T and HR. In addition the cost drivers for this set of costs are consistent along with the allocation methodology.

## 2 Return on Capital

The Return on Capital component of the revenue cap (representing a reasonable rate of return) can be separated into three distinct cost groups on the basis of asset type. These cost groups are:

- **network return on capital** – the return on those system assets employed in the provision of network connection and distribution services to customers (including any applicable tax);
- **common services return on capital** – the return on those system assets associated with the provision of common services; and



- **other return on capital** – the return on non-system assets (for example, fleet, computers and buildings) and working capital, which are employed to provide regulated services to customers.

Return on capital costs are allocated to each cost group in proportion to the value of assets.

### 3 Regulatory Depreciation

The regulatory depreciation allowance can also be separated into the three cost groups as with the return on capital allowance, as follows:

- **network assets** – the depreciation of those assets employed for the provision of network connection and distribution services to customers;
- **common services assets** – the depreciation of those assets associated with the provision of common services activities; and
- **other assets** – the depreciation of the non-system assets (for example fleet, computers and buildings) which are employed to provide regulated services to customers.

Regulatory depreciation costs are allocated to each cost group in proportion to the value of assets.

### 4 Tax Allowance

The benchmark tax liability as specified in the PTRM is allocated to each of the three building block components of operating expenditure, return on capital and regulatory depreciation in the ratio of their respective allocated revenue amounts. Within each of these regulatory cost components there are a number of cost groups. The allotted tax allowance will be allocated to each cost group on the basis of their respective share of the total revenue and further allocated to users of asset classes on the same basis for the respective cost group.



## 5 DCOS Cost Categories

Table 1 below sets out the translation of these cost groups which relate to the components of the revenue cap into DCOS cost categories that are homogeneous in relation to cause and allocation approach. Table 2 in the next section sets out the allocators that are used to distribute these costs across the relevant customer groups.

**Table 1 Cost Groups to DCOS Cost Categories**

| Revenue Cap Components               | Cost Groups                        | DCOS Cost Categories |                 |            |
|--------------------------------------|------------------------------------|----------------------|-----------------|------------|
|                                      |                                    | Network              | Common Services | Non-system |
| Operating Expenditure<br>Section 1   | Direct – Network                   | ✓                    |                 |            |
|                                      | On-costs                           | ✓                    |                 |            |
|                                      | Direct – Common Services           |                      | ✓               |            |
|                                      | Overheads and other Operating      |                      |                 | ✓          |
| Return on Capital<br>Section 2       | Network Return on Capital          | ✓                    |                 |            |
|                                      | Common Services Return on Capital  |                      | ✓               |            |
|                                      | Non-System Return on Capital       |                      |                 | ✓          |
| Regulatory Depreciation<br>Section 3 | Network Assets Depreciation        | ✓                    |                 |            |
|                                      | Common Services Asset Depreciation |                      | ✓               |            |
|                                      | Non-system Assets Depreciation     |                      |                 | ✓          |
| Tax Allowance<br>Section 4           | Benchmark Tax Liability            | ✓                    | ✓               | ✓          |

## 6 Allocation of DCOS Cost Categories to Tariff Classes

There are 7 tariff classes (for *standard control services*) to which network costs are allocated. These groups are:

- a) ICCs;
- b) CAC – 33kV;
- c) CAC – 11kV Bus;
- d) CAC – 11kV Line;
- e) SAC - demand;
- f) SAC – non-demand; and
- g) EGs.

The purpose of the cost allocation process is to develop a 'cost for the provision of network connection and access services' for each of these tariff classes based on appropriate cost drivers and/or cost allocations.



## 6.1 Tariff Cost Allocators

There is a range of cost allocators that can be used in a DCOS model such as those used by ENERGEX. The selection of the appropriate allocator is based on the ability of that allocator to reflect the fundamental cost driver. The range of possible allocators includes:

- a) number of customers;
- b) anytime energy;
- c) period energy (time of use);
- d) anytime demand;
- e) period demand (time of use); and
- f) coincident demand.

ENERGEX has adopted the allocators listed below in the cost allocation model for the following reasons:

- **Number of customers** – this allocator is appropriate for those costs that are dependent upon or driven by the number of connected customers. ENERGEX has a number of costs that are customer number based. Indeed a significant proportion of the overhead costs of the business are driven by the number of staff and systems required to serve the customer base;
- **Anytime energy** – this has been used as an allocator for those costs that are related to the size of the customer but not specifically the demand of that customer on the network. Those costs that cannot be directly related to a product or service are recovered through the use of energy prices (for example, market fees, and ancillary services fees); and
- **Anytime maximum demand (ATMD)** – this method of allocation has been used for the shared system costs. Whilst the ideal cost allocation mechanism would be based on a real-time model which replicates these network location-specific demands, such an approach is not achievable at present. ATMD provides a simple and reasonable basis for apportioning system usage related costs given the present availability of data. It reflects the fact that demand is the primary driver of shared network costs whilst overcoming the limitations that other allocators have in recognising the variations across the network. The demand forecast approved in the Final Determination is a coincident system peak forecast demand. Whereas the total of the ATMDs used in DCOS will not sum to the coincident system peak due to diversity in the network.



## 6.2 Implemented Cost Allocations

Table 2 and Figure 1 below set out the DCOS cost categories, the cost driver and the tariff cost allocation process adopted in accordance with the previously established principles. Within each of the groups of network customers, there are a number of distribution network users with a range of network tariffs. The allocation of costs within a group to the network customers is on the same basis as for the general allocation of costs to the groups as described in Table 2.

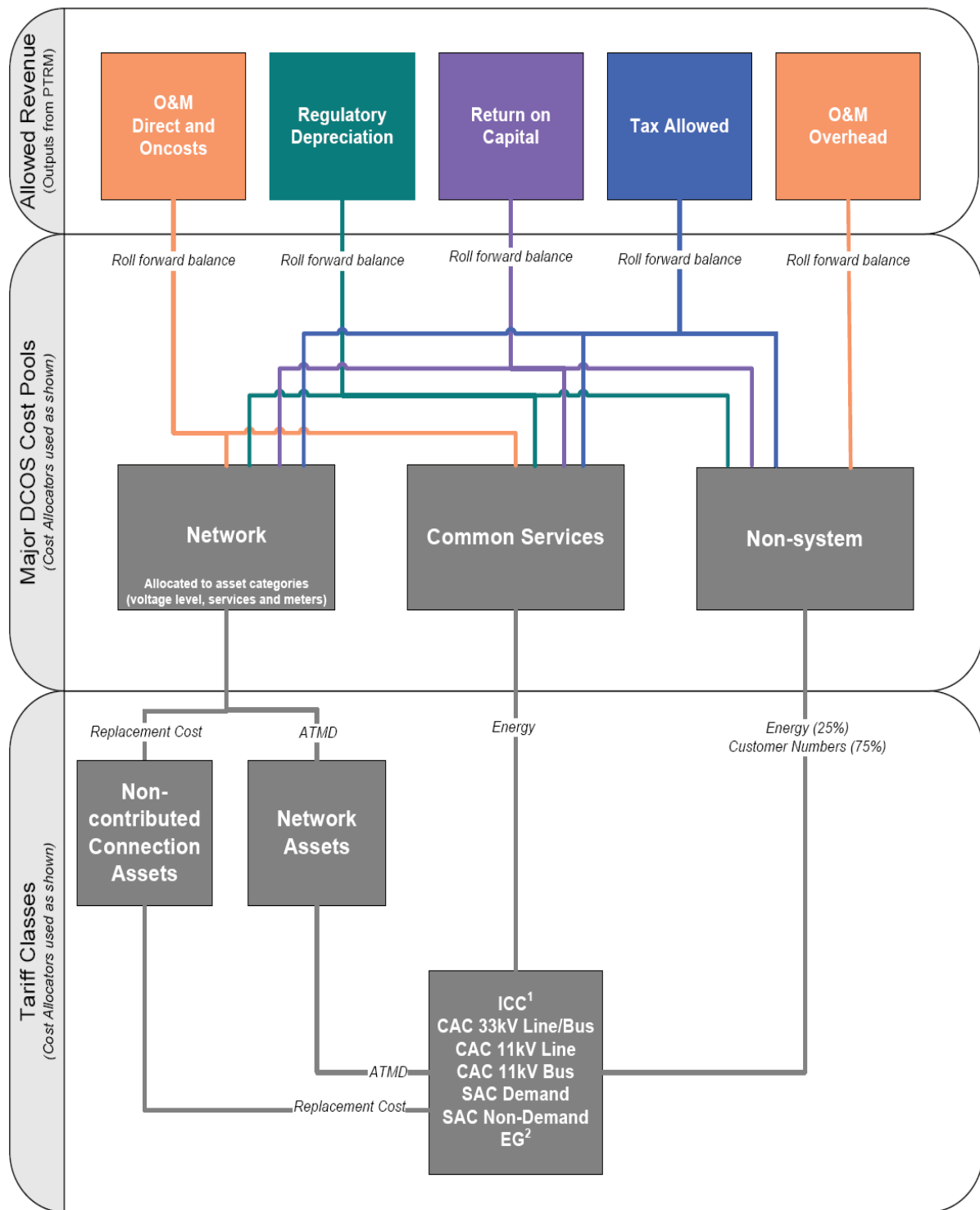
**Table 2 Allocation of Costs to User Groups**

| DCOS Cost Category                              | Cost Driver                                                                                                            | Tariff Cost Allocation Process                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network O&M                                     | Connection capacities<br>Demand on relevant parts of the network<br>Number of connections                              | Costs apportioned to asset categories (asset groups at voltage level) on the basis of direct costs (direct – network operating expenditure). On-costs are apportioned consistent with direct network operating expenditure. The costs are then allocated to tariff classes on the basis of the use of those assets through direct allocation for connection assets and ATMD for shared assets.                                                                                                   |
| Network regulatory depreciation                 | Non-contributed connection capacities <sup>1</sup><br>Demand on relevant parts of the network<br>Number of connections | Costs apportioned to assets (in DCOS categories) by voltage level proportional to the replacement costs. Costs are then allocated to tariff classes on the basis of the use of those assets through direct allocation for non-contributed connection assets and ATMD for shared assets.                                                                                                                                                                                                          |
| Network return on capital                       | Non-contributed connection capacities <sup>1</sup><br>Demand on relevant parts of the network<br>Number of connections | Costs apportioned to assets (in DCOS categories) by voltage level proportional to the replacement costs. Costs are then allocated to tariff classes on the basis of the use of those assets through direct allocation for non-contributed connection assets and ATMD for shared assets.                                                                                                                                                                                                          |
| Common services                                 | System control and stability<br>No unique cost driver                                                                  | Costs allocated to all customers using energy as the allocator.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Non-system (Overhead and other operating costs) | Number of customers connected<br>Service requirements of customers                                                     | These costs are associated with the number of connected customers and their expectations/service requirements. A hybrid allocation of a percentage of costs per customer (75%) and a percentage of costs on a c/kW.h (25%) is adopted. These weights reflect that the number of customers is the primary driver of the other costs.<br><br>EGs will be allocated a fixed dollar (\$) cost depending on size and type of connection. These costs reflect the operational service required by EGs. |
| Capital Contributions                           | Number of customers connected<br>Service requirements of customers                                                     | These costs are associated with the number of connected customers and their expectations/service requirements. A hybrid allocation of a percentage of costs per SAC customer (75%) and a percentage of costs on a c/kW.h (25%) is adopted.                                                                                                                                                                                                                                                       |

<sup>1</sup> Contributed connection assets are not used in the allocation of regulatory depreciation or return on capital



**Figure 1 Tariff Cost Allocation Process Flowchart**



**Notes:**

- 1 For ICC customers the dedicated cost allocation includes the cost of their connection assets plus their proportion of the shared network.
- 2 For EG customers the non-system costs will be allocated on a fixed dollar amount based on the size and type of connection. These costs reflect the additional operational services required by EGs



### **6.3 Variances from allocated cost based tariffs**

ENERGEX develops and applies network tariffs based on the distributed cost model. For each tariff class, the costs outlined above are recovered through a combination of fixed charges, demand charges and/or volume charges. The network pricing methodology applied to each of those groups has precluded any possible bypass challenge on the basis that the network tariff is efficient and an alternative electricity service can not be sourced at a lower economic value. Providing it is consistent with the NER, ENERGEX may negotiate a tariff other than the tariff calculated using the cost allocated methodology where it can be demonstrated that the cost-based network tariff is not efficient, as an economic bypass opportunity exists or an alternative electricity service could be utilised.



## **Appendix 4**

### **Reclassification of customers to tariff classes (confidential)**

## Appendix 4

**CONFIDENTIAL**

## Reclassification of customers to tariff classes

**Table 1      Reclassifications for Customers in 2010-11**[illegible]



## **Appendix 5**

### **Street light Classification**

## Appendix 5

### Street light classification

Street lights are included in the unmetered street lighting tariffs, according to the nominated Wattage Reference Codes as detailed below.

**Table 1 Street lights – minor classification**

| Street Light Type | Wattage Reference Codes       |
|-------------------|-------------------------------|
| Mercury Vapour    | M50, M70, M80, M100, M125     |
| Mercury Halide    | H35, H70, H75                 |
| Sodium Vapour     | L35, L90, L135, S35, S50, S70 |
| Incandescent      | All                           |
| Fluorescent       | All                           |

Also includes any other non-standard or obsolete public lights that would be replaced with any of the above ENERGEX standard minor public lights in accordance with ENERGEX policy.

**Table 2 Street lights – major classification**

| Street Light Type | Wattage Reference Codes                                  |
|-------------------|----------------------------------------------------------|
| Mercury Vapour    | M150, M175, M250, M400, M500, M700, M1000                |
| Mercury Halide    | H100, H150, H250, H400, H1000                            |
| Sodium Vapour     | S100, S150, SM160, S250, S360, S400, S2X400, S800, S1000 |

Also includes any other non-standard or obsolete public lights that would be replaced with any of the above ENERGEX standard minor public lights in accordance with ENERGEX policy.



## **Appendix 6**

### **In-period Price Review Process**

## Appendix 6

# In-period price review process

### 1 Standard Control – Network tariffs

ENERGEX calculates and proposes the network tariffs for existing customers at the beginning of the regulatory year.

Within a regulatory year there may be a requirement to include a new site specific charge due to:

- a new large customer approaching ENERGEX with connection requirements, or
- an existing customer requiring a substantial change to their current connection.

The procedure below describes the proposed process for the approval of a new or revised site specific price for large customers.

Having identified the need for a change to site specific prices, ENERGEX will submit a proposal to the AER for consideration. The proposal will include:

- a) The nature and details of the proposed connection;
- b) The methodology used for calculation of the proposed price based on the pricing principles approved in the current pricing proposal;
- c) Details of the proposed price and the commencement date for the site specific charge;  
and
- d) Supporting documentation.

ENERGEX will supply all additional information as requested by the AER, to allow them to approve the submitted price.

It is envisaged that the AER will notify ENERGEX in writing of their decision within 30 days.

Following notification of the AER's decision, ENERGEX will update the large customer price schedules and systems accordingly and notify the customer and their retailer of any change.



## **2 Alternative Control – Fee based services**

ENERGEX aims to provide services that meet both statutory obligations and community expectations. Accordingly, from time to time it may be necessary to introduce a new fee based service. The procedure below describes the proposed process.

Having identified the need for a change to the service listing, ENERGEX will submit a proposal to the AER for consideration. The proposal will include:

- a) The nature of the proposed amendment;
- b) An explanatory statement of the reasons for the proposed amendment;
- c) The proposed price for the service; and
- d) Supporting documentation.

ENERGEX will supply all additional information as requested by the AER, to allow them to approve the new fee based service.

It is envisaged that the AER will notify ENERGEX in writing of their decision within 30 days.

Following notification of the AER's decision, ENERGEX will update the fee based price schedules and systems accordingly, publish any amendments on the ENERGEX website and notify retailers of any changes.



## **Appendix 7**

### **Capital Contributions Policy**



**ENERGEN Limited**

**ABN 40 078 849 055**

# **CAPITAL CONTRIBUTIONS POLICY**

**MAY 2010**

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## 1. INTRODUCTION

Capital contributions apply primarily to newly connecting Standard Asset Customers (SACs), and are sought as prepayment for a revenue shortfall in the case of an uneconomic connection. An uneconomic connection is defined as one where the average distribution prices for the relevant network price category would not be sufficient to recover the full cost of the connection.

For new SACs wishing to access the standard DUOS tariffs, some level of contribution to the costs of connection may be necessary. ENERGEX determines the level of capital contribution with reference to the following over-arching principles:

- Ensuring equitable treatment of existing customers; and
- Sending appropriate pricing signals to potential customers that reflect the true cost of network expansion.

Connection costs for Individually Calculated Customers (ICCs), Connection Asset Customers (CACs) and Embedded Generators (EG) are specifically included in their site-specific network charges.

ENERGEX received confirmation from the AER on 18 December 2009 that the capital contributions policy previously approved by the Queensland Competition Authority (QCA) satisfied the requirements under the Rules, Clause 11.16.10. The AER confirmed that it was not necessary for ENERGEX to revise or republish the policy.

However, ENERGEX has amended the policy to ensure consistency in terminology for the 2010-15 regulatory control period and has removed obsolete (historical) sections where appropriate to provide a more concise document. ENERGEX has ensured that the principles contained in this capital contributions policy proposed for the 2010-15 regulatory control period are consistent with the previous approach approved by the QCA.

## **2. CAPITAL CONTRIBUTIONS POLICY**

This is ENERGETEX's policy for seeking capital contributions from new customers connecting to the electricity network and is primarily targeted at SACs. The policy is consistent with the pricing principles outlined in ENERGETEX's Pricing Proposal, specifically, that the capital contributions policy:

- Ensures the equitable treatment of existing customers; and
- Sends appropriate pricing signals to potential customers that reflect the true cost of network expansion.

### **2.1 What are Capital Contributions**

Capital contributions are made by a customer to the network operator where the network operator considers that it would not be able to fully recover the costs of delivering services to that customer through average annual prices, ie the connection is uneconomic. When this assessment is made, a capital contribution towards connection investment is sought as a prepayment for the expected revenue shortfall.

### **2.2 Principles for Setting Capital Contribution**

The level of a particular capital contribution is determined by reference to the following objectives:

- To meet the economic efficiency objective, capital contributions should only cover any shortfall between the present value of distribution charges expected to be paid by the new customer over the life of the assets and the incremental cost of connecting that customer. This approach ensures that existing customers are no worse off following the connection of a new user because the expected network revenue from the new customer (in the form of additional charges and/or capital contributions) will cover the incremental cost of supply; and
- To meet the equity objective, it is reasonable to expect each customer, in addition to their incremental costs of connection to make some contribution to shared assets.

The test to be applied to new SAC customer connections incorporates both economic efficiency and equity components.

#### **2.2.1 ECONOMIC EFFICIENCY**

The economic efficiency test is based on the principle that existing customers will be no worse-off following the introduction of a new user, that is, the expected network revenue from the new user at least covers the incremental cost of supply.

Operationally, therefore, if the present value of future DUOS revenue from the new user is greater than the incremental costs of supplying distribution services to the new user, then all existing customers will be better off. If the expected revenues are insufficient to cover these incremental costs (and the entrant wants to access standard network tariffs), then a contribution will be required. If no contribution was sought from this customer, and they were allowed to access standard network tariffs, average DUOS charges for all customers on these tariffs would increase. Therefore, the minimum capital contribution will be calculated by deducting the present value of expected future DUOS revenues from total incremental cost.

### 2.2.2 EQUITY

Equity considerations suggest that new customers should, in addition to meeting incremental costs, contribute to the costs of the existing shared network. These costs, which in economic terms are defined as joint production costs<sup>1</sup>, do not have a separable cost function and therefore cannot be specifically determined by varying the relative proportions of the services provided. Consequently, there is no objective way of attributing joint production costs to the new distribution service. Any method used to allocate or attribute shared network costs to new customers will result in one of a broad range of possible outcomes. The acceptable range of outcomes is constrained by the avoidable/stand-alone cost rule that is explicitly employed in ENERGEX's pricing principles. Nevertheless, for a new customer this range could be considerable.

In determining a reasonable level of shared network contribution for a new connection, ENERGEX has had regard to the range of contributions made by existing customers. Contributions to shared network costs will vary significantly across the network. However, it is proposed that a standard contribution of 10% of expected DUOS revenues will be applied when determining capital contribution levels for SAC customers because this approximates the minimum contribution to shared network costs that is currently made by SAC customers.

The following section outlines the approach that ENERGEX has employed in determining the level of contribution new customers are required to make.

## 3. METHOD OF CALCULATING CAPITAL CONTRIBUTIONS

The method used to calculate capital contributions is based on the following:

- Forecast DUOS revenue based on DUOS charges applicable to the relevant tariff class. The DUOS rates to be used in the calculation would be the tariffs as published annually in ENERGEX's Pricing Proposal to the AER;
- A discount rate based on ENERGEX's approved regulatory WACC converted to pre-tax terms using the estimated average effective tax rate for the regulatory period;
- Twenty year discount periods for both domestic and commercial customers as this represents a reasonable expectation with regard to the average life of the assets and the period of time before customers are expected to seek capacity enhancements;
- A 10% reduction in the discounted DUOS charges to account for the contribution to the shared network;
- Calculating the incremental costs of connection will include two elements (both of which are calculated before ENERGEX applies GST);
  - The actual cost of connection (the capital cost of the network connection assets including overhead service and metering equipment); and
  - Additional costs, both capital and operating, incurred in the shared (upstream) network as a direct result of the new connection.

For most SAC customers, identifying specific upstream network costs is extremely difficult and is generally expected to be immaterial. However, where they can be clearly identified and are considered material, they will be included in the calculation of total incremental costs.

---

<sup>1</sup> Reflecting economies of scope.

Calculation of a new customer's capital contribution level can be expressed mathematically as follows:

$$CC = ICCS + ICSN - [IR(n=20) - SNC(10\%)]$$

Where:

|          |   |                                                                                                  |
|----------|---|--------------------------------------------------------------------------------------------------|
| CC       | = | Capital Contribution                                                                             |
| ICCS     | = | Customer specific incremental costs                                                              |
| ICSN     | = | Incremental costs in the upstream (shared) network directly attributable to the new connection   |
| IR(n=20) | = | Present value of a 20 year revenue stream directly attributable to the new connection            |
| SNC(10%) | = | A 10% attribution of Incremental revenue (IR(n=20)) to the costs of the existing shared network. |

## 4. SHARING CAPITAL CONTRIBUTIONS

Following a capital contribution for a network extension from the original connecting customer, new customers may connect to that section. ENERGEX's policy for sharing capital contributions is outlined in this section.

### 4.1 Method of Sharing

Past contributions are not refundable. However, new contributions may be shared with past customers – that is, the amounts contributed by additional customers become the basis of any payment to the customer who paid the initial contribution.

ENERGEX's policies relating to the sharing of capital contributions are outlined below:

- Initial contribution levels are reduced to zero over a period of five years (20% reduction annually from date supply is available). The impracticality of administering such a scheme on an indefinite basis requires that some limit be placed on the period over which capital contributions can attract refunds from future customers.  
IPART (April 2002) found that most interested parties considered a period in the range of 5-7 years reasonable. While IPART decided on a 7-year period, it restricted the application of capital contributions to rural and large load customers. ENERGEX employs no such restriction on the application of capital contribution and hence a shorter period is applicable given the greater turnover of smaller urban properties;
- Subsequent contributions to the initial extension should be calculated proportionate to customer size based on forecast DUOS revenue from DUOS charges applicable to the relevant tariff class;
- No sharing of contributions is permitted for additional customers connected to an extension supplied under a Rural Subsidy Scheme; and
- Amounts less than \$100 will not be refunded.

## 5. IN KIND CAPITAL CONTRIBUTIONS

In some cases customers connecting to ENERGEX's network will provide In Kind capital contributions (non-cash). This section outlines ENERGEX's treatment of In Kind capital contributions.

## 5.1 Individual Customers

For an individual customer connection providing In Kind (non-cash) capital contributions the following approach applies:

- Where the total project cost (as determined by ENERGEX) is less than the incremental revenue threshold, ENERGEX will repay the total cost to the customer; and
- Where the total project cost (as determined by ENERGEX) is greater than the incremental revenue threshold, an amount equal to the threshold will be repaid to the customer with the residual recorded as an In Kind capital contribution to be netted-off against the allowed network revenue limit.

## 5.2 New Developments

ENERGEX also receives In Kind capital contributions from developers for multiple connections associated with new developments.

### 5.2.1 BACKGROUND

In July 2006, an amended capital contributions policy for new developments was introduced. This replaced the previous policy which had been in place since the late 1980's. ENERGEX implemented a phased introduction for this new policy over the period 2006-07 to 2008-09. This timeframe was introduced for equity reasons and to provide developers with time to adjust to the new policy (given lead times in the industry from project conception to completion of approximately two to three years). The transitional period for this new policy has now concluded.

### 5.2.2 APPROACH

The current approach for new developments is that developers will be required to pay for all costs (inside the development and upstream augmentations) associated with supplying electricity to lots within the development. From July 2006, ENERGEX's capital contributions policy for In Kind contributions from developments has incorporated:

- The phased introduction of contestability for the procurement of major electricity reticulation materials by developers;
- The introduction of an ENERGEX approved standard for design, specification and quality levels for the procurement of major electricity reticulation materials purchased by developers and their sub-contractors;
- The phased removal of the contributed value per lot for electricity reticulation materials supplied by ENERGEX to developers from July 2006 to July 2009;
- ENERGEX will contribute to the costs for the following works:
  - Feeder and shared network augmentation outside of the development that has specifically been required by ENERGEX that is above what is required to supply the development (for example tie cables for reliability purposes); and/or
  - Backbone network augmentation within the development requested by ENERGEX that is above that required to supply the development.
  - ENERGEX's contribution to a development will not exceed the total costs of the additional works requested by ENERGEX.
- The developer will be responsible for the full cost required to service the development.

## 6. SUMMARY

In summary, ENERGETIC's capital contributions policy includes:

- Calculation of any required capital contribution by deducting the present value of expected future revenues from the incremental cost of the connection;
- The use of expected DUOS revenues for SAC customers as the basis for calculating incremental revenue;
- Discounting expected future revenues using ENERGETIC's approved WACC expressed in pre-tax, real terms;
- Allocating 10% of the forecast revenue stream as a contribution to the costs of the existing shared network; and
- Taking account of any specific upstream impacts in the calculation of incremental costs, where appropriate.